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The Stability Gap as a Trajectory Problem: A Bent Descent Path, Its Overshoot, and a Single Gate

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation exposes a failure this view hides: the *stability gap*, a sharp but transient drop in past-task accuracy at the start of every new task. The gap persists even when the optimiser descends the exact joint loss over all tasks, so its cause lies in the optimisation trajectory rather than the objective. This thesis explains the gap on two levels. Learning both tasks has a fixed price: the loss on past tasks must rise to its level at the joint optimum. An envelope-theorem argument shows that a reference path exists along which this rise is monotone. Steepest descent after an abrupt task switch leaves that path: it bows out of the old task's basin and returns from outside, a deterministic arc that survives exact gradients (level one, the trajectory problem). Real optimisers also overshoot whatever path they follow: an imbalanced first step, gradient noise from a finite buffer, and momentum each push the iterate beyond it (level two, the overshoot problem). Written in a unified update rule, every replay method counters these problems with the same tool, a *gate* on the new-task gradient, while the replay gradient restores at full strength. We build and compare the two pure gate designs. The λ-*curriculum* is a scalar feedback gate: it ramps the new-task loss weight up from zero, so the optimiser tracks the moving valley floor of the reference path. *Asymmetric preconditioned experience replay* is a directional feedforward gate: it filters the new-task gradient through past-task curvature, passing it where the past task is flat and braking it where the past task is steep. On rotated MNIST the two levels separate cleanly. Removing the overshoot drivers leaves the arc; placing the optimiser on the reference path leaves the overshoot; doing both removes the gap while the fixed price remains. The curriculum needs momentum for accuracy, not for the gap: momentum acts as a variance reducer that closes the residual gap and as an accelerator that restores plasticity. The directional gate lowers the gap monotonically as its filter strengthens, at zero plasticity cost, but it fails through blind spots of its sampled curvature metric and its protection unravels under momentum. The two designs therefore trace a stability–plasticity frontier rather than a ranking. On CIFAR-10 with a ResNet-18 the curriculum cuts gap depth several-fold below vanilla experience replay at matched final accuracy, under two normalisation schemes and on corruptions it was never tuned on.

Contents

1 Introduction

Deep neural networks learn powerful representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [18, 21], is a major hurdle in machine learning. Retraining from scratch solves the issue, but it is too expensive for systems that need to adapt continuously. Continual learning (CL) studies how to learn from a sequence of tasks without that cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [19].

Replay, the family this thesis studies, stores a small buffer of past examples and trains on them alongside the new data. Replay methods reduce long-term forgetting well, but until recently they were evaluated only at task boundaries, which hides what happens within a task. Lange et al. [16] evaluated the network after every update and identified the *stability gap*: a severe but transient collapse in past-task accuracy that occurs immediately when a new task begins, followed by a slow recovery. This matters for two reasons. A deployed system is queried throughout training, so a model caught mid-transition commits errors its final checkpoint never would. And because the model eventually recovers, past knowledge is disrupted rather than lost, which means the disruption itself is the target for repair.

The discovery raised a natural question: would a better approximation of the joint loss over all tasks remove the gap? Hess et al. [11] showed that the gap persists even under incremental joint training, where the model has access to all data from all tasks at every step. A perfect objective still produces transient forgetting. The problem therefore lies in *how* the optimisation proceeds, and this thesis locates it on two levels: the descent path itself is bent, and real optimisers overshoot whatever path they follow.

Level one: the path bows. Consider the geometry of the transition. Figure 1.1 visualises a switch from a learned task (Task A) to a new task (Task B), a conceptual two-dimensional reduction of a high-dimensional surface. Our depiction borrows the loss-landscape lens of Kao et al. [14]. The network starts at the converged minimum A and must reach the joint minimum $A \cap B$. Steepest descent on the joint loss follows the green curve: because the steepest direction differs from the direction toward the joint minimum whenever curvature is anisotropic, the trajectory bows outward, leaves Task A's basin, and returns from outside [14]. Along this arc the Task-A loss rises above its final level and then comes back down. The arc is deterministic: it appears with exact gradients and vanishing step size, so it survives every improvement in how carefully the path is followed. Its signature in the accuracy curve is a hanging tail: accuracy first drops and then recovers only slowly, staying depressed while the iterate walks the long way around.

What must be paid, and what is avoidable. The arc is a defect of the path, and better paths exist. Reaching the joint optimum has one unavoidable cost: the past-task loss must end at its (higher) value at the joint minimum. That rise, however, can be monotone. Section 4 constructs a reference path along which the past-task loss only grows toward its final level (the dashed line in Figure 1.1, which stays inside Task A's basin rather than bowing out), using an envelope-theorem argument, and the mode-connectivity literature confirms that low-loss routes between such solutions exist in practice [6, 20]. This separates the permanent stability–plasticity price from the stability gap: everything the accuracy curve does *below its eventual settled level and back* is a failure to follow a monotone path, and is avoidable in principle.

Level two: optimisers overshoot the path they follow. The arc is what an idealised gradient flow traces; a real optimiser overshoots it. At the boundary the objective switches in a single step from the replay loss alone to the sum of replay and current-task losses, so the minimum the parameters chase teleports from θ_A^* to θ_{joint}^* while the parameters stand still. This *landscape teleportation* leaves the iterate maximally imbalanced: the replay gradient is near zero because the optimiser has converged on Task A, while the new-task gradient is at its largest because Task B is unlearned

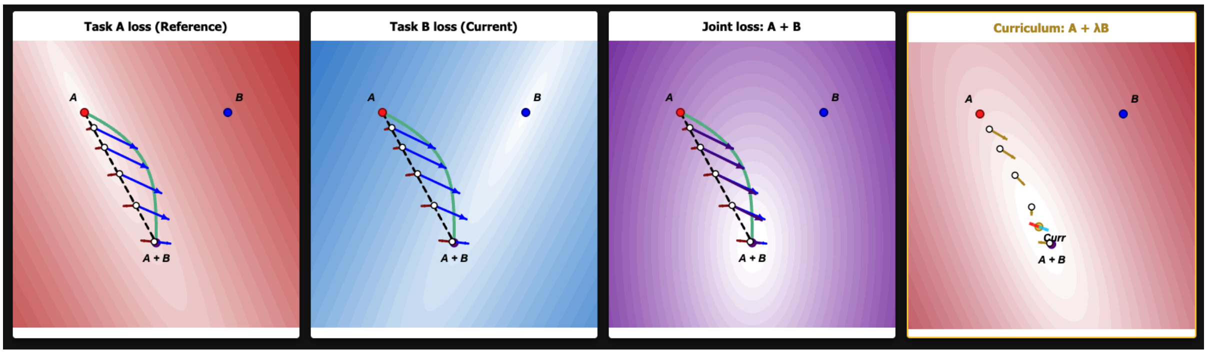

Figure 1.1: Conceptual visualisation of the optimisation landscape at a task transition, shown over a shared two-dimensional parameter space. Task A (red) and Task B (blue) are the single-task losses with minima at A and B; the joint loss $A+B$ (purple) has its minimum at $A \cap B$. The green curve is the steepest-descent trajectory of the joint loss: it bows away from Task A's basin, crosses a region of higher Task-A loss, and approaches $A \cap B$ from outside. The dashed black line is the direct chord from A to $A \cap B$, which stays inside Task A's basin: it is the monotone reference path of Section 4, along which the Task-A loss only rises toward its joint value. This arc is the trajectory problem of Section 3.2; the overshoot drivers of Section 3.3 push the iterate even further off whichever path it follows. The fourth panel shows the λ-curriculum loss $A+\lambda B$: a small weight λ on Task B reshapes the basin into a valley that runs continuously from A toward the joint minimum, so the optimum moves along the corridor and the optimiser can follow it. Arrows are normalised to a fixed length and show gradient direction only.

[16]. Taken as a finite step, this imbalanced push carries the iterate far out of the old basin in the first few iterations. Its signature in accuracy curve is a steep drop followed by a steep recovery. Two sources of variance widen and prolong the excursion: the finite replay buffer estimates the restoring gradient from a small sample [1], and each minibatch re-estimates the gradients afresh, so the iterate jitters around the path and settles back slowly. This overshoot acts on any path, the bowed one and the reference alike.

One tool: gate the new-task gradient. Sequential replay methods share a single update structure, a weighted combination of the current-task and replay gradients [22]. Section 3.4 generalises this observation: every replay method applies a *gate* to the new-task gradient and lets the replay gradient restore at full strength,

$$d \;=\; G\, g_{\text{cur}} \;+\; g_{\text{rep}}.$$

Methods differ in two design choices. The gate's *domain* can be scalar (attenuate the whole push) or directional (attenuate per direction). Its *control* can be feedback (react to a live signal) or feedforward (commit to a schedule or a precomputed filter). This thesis builds and compares the two pure designs.

The first is the **λ-curriculum**, a scalar gate. Instead of switching abruptly to the joint loss, it ramps the new-task weight from 0 to 1,

$$\begin{aligned} L_\lambda(t) &= L_{\text{replay}} + \lambda(t)\, L_{\text{current}}, \\ \lambda(t) &= \operatorname{clip}\!\left(\tfrac{t}{N}, 0, 1\right), \end{aligned} \tag{1.1}$$

so the minimiser of the objective moves along a continuous path from θ_A^* to θ_{joint}^* and the model tracks a moving valley floor (fourth panel of Figure 1.1). Its adaptive variant sets λ from the ratio of the replay and new-task gradient norms, a live damage signal, which makes it a scalar *feedback* gate. The second is **asymmetric preconditioned experience replay** (asymmetric PER), a directional *feedforward* gate. It filters the new-task gradient through the curvature of the replay loss,

$$d \;=\; \delta\,(F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}} \;+\; g_{\text{rep}}, \tag{1.2}$$

where F_{rep} is the replay Fisher information and δ sets the filter strength. The push passes at full gain where the past task is flat and is braked where the past task is steep, while the restoring replay gradient stays untouched.

Contributions. This thesis makes four contributions.

1. **A two-level account of the stability gap.** The gap separates into a deterministic arc of the descent path and overshoot of the followed path, with the permanent stability–plasticity cost split off by a monotone reference path (Sections 3 and 4). Controlled experiments confirm the separation: removing the overshoot drivers leaves the arc, fixing the path leaves the overshoot, and doing both removes the gap.

2. **A unified gate view of replay methods**, which places existing methods and the two designs of this thesis in one two-axis taxonomy and turns the taxonomy's predicted failure modes into testable hypotheses (Section 3.4).

3. **The λ-curriculum**, a scalar feedback gate with an $O(1/N)$ tracking guarantee, an adaptive schedule, a $\lambda_{\min}$ plasticity floor, and a characterisation of its interplay with momentum.

4. **Asymmetric PER**, a directional feedforward gate, with a damping sweep that maps its stability–plasticity trade-off and identifies its failure modes: blind spots of the sampled curvature metric and the unravelling of per-step protection under momentum.

Section 2 reviews continual learning, experience replay, the stability gap, and the unified update that underlies the gate view. Section 3 develops the two-level account and the gate taxonomy. Section 4 constructs the monotone reference path and proves the tracking bound. Section 5 turns the account into four research questions, and Section 6 describes the testbeds, metrics, and conditions that answer them. Section 7 reports the experiments on rotated MNIST and CIFAR-10, and Section 8 weighs the evidence and states the limitations.

2 Background and Related Work

2.1 Continual Learning and Experience Replay

Methods against catastrophic forgetting fall into a small number of families: replay, parameter and functional regularisation, optimisation-based methods, context-dependent processing, and template-based classification [24]. This thesis studies replay, and in particular its simplest member, Experience Replay (ER) [3]. At each step ER draws a replay mini-batch from a buffer of stored past examples, combines it with the current-task mini-batch, and takes a single optimiser step on the summed loss. We populate the buffer by reservoir sampling [25], one of several buffer-management policies, so that every sample seen so far occupies a slot with equal probability. During the first task the buffer is empty by construction, so ER reduces to plain SGD, and the stability gap becomes measurable from the second task onward.

This simplicity is why ER remains competitive: joint training with a small episodic memory often outperforms purpose-built methods and scales gracefully to longer sequences [3]. It also makes ER the clean substrate for this thesis, because its update contains exactly two ingredients, the new-task gradient and the replay gradient, and every intervention we study acts on how the two are combined.

A finite buffer is an imperfect estimator of the true past-task distribution. Aljundi et al. [1] frame buffer selection as approximating the feasibility region defined by the full past data; reservoir sampling is one policy within that frame. The replay gradient computed on a small buffer therefore differs in direction and magnitude from the gradient on all past data, at every step and independent of any schedule. Section 3.3 takes up how this sampling noise feeds the overshoot at a task boundary.

2.2 The Stability Gap

Lange et al. [16] identified the stability gap using continual evaluation, measuring past-task accuracy after every update rather than only at task boundaries. In this view a new task begins with substantial but transient forgetting that the model subsequently recovers from, and they quantify it with worst-case metrics, mainly the minimum past-task accuracy reached during training, rather than with end-of-task averages alone. The effect is not confined to one method: they observe it across experience replay, constraint-based replay, knowledge distillation, and parameter regularisation. It also persists well beyond the standard benchmarks, appearing in joint incremental learning of homogeneous tasks [13] and in continual pre-training of large language models [7], and several mitigations have been proposed [9, 23].

Two findings in this literature anchor the account of Section 3. First, Lange et al. [16] trace the initial drop to a gradient imbalance at the boundary: the new-task gradient dominates the first updates, so the optimiser trades stability for plasticity precisely when the past task is most exposed. Second, the gap is a property of the trajectory rather than the objective. Hess et al. [11] show it persists under training on the exact joint loss with full access to all past data, and Kao et al. [14] demonstrate the geometric mechanism analytically: steepest descent from an old-task minimum leaves the old basin even with perfect joint gradients, because the steepest direction and the direction to the joint minimum disagree. A perfect objective therefore fails to prevent the gap, and the descent geometry itself must be addressed.

2.3 One Update, Many Methods

A second class of replay-adjacent methods constrains the update at the task boundary using information about past tasks. GEM stores past-task gradients and projects the new-task gradient onto the nearest direction that leaves every stored loss unchanged or lower [17]. A-GEM replaces the per-task constraints with a single averaged past-task gradient, reducing the projection to one cheap operation [2]. EWC adds a quadratic penalty around the past optimum weighted by a curvature estimate [15], and natural-gradient approaches precondition the update with past-task curvature so that steps avoid directions the old task depends on [14].

On the surface these methods look unrelated, but Sodhani et al. [22] show the replay-based ones share a single update. Every step combines two gradients, the current task's and the replay buffer's, under two scalar weights,

$$\theta \leftarrow \theta - \eta \left(\alpha_1 \nabla L_{\text{current}} + \alpha_2 \nabla L_{\text{replay}}\right), \qquad (2.1)$$

so a method reduces to a rule for choosing α_1 and α_2. Plain ER fixes both to one. GEM and A-GEM hold $\alpha_1 = 1$ and raise the replay weight α_2 only when the new-task and stored gradients conflict, by just enough to cancel the interference; A-GEM's projection is therefore the addition of scaled losses. MEGA [8] likewise holds $\alpha_1 = 1$ but sets the replay weight from a loss ratio, $\alpha_2 = L_{\text{replay}}/L_{\text{current}}$, leaning on memory once the new task is already fit.

The same reading applies to the λ-curriculum of this thesis. It fixes $\alpha_2 \equiv 1$ and turns the other knob, ramping the current-task weight $\alpha_1 = \lambda(t)$ up from zero. It belongs to the same family as A-GEM and MEGA; what distinguishes it is which weight it moves and what signal drives the weight. The adaptive variant (Section 6.2.2) is the mirror image of MEGA: MEGA raises the *replay* weight from a *loss* ratio to recruit memory once the new task is learned, while the adaptive curriculum holds the *new task* back with a *gradient-norm* ratio, letting it in only as the replay gradient reasserts itself. Section 3.4 generalises Equation (2.1) from scalar weights to operators and organises all of these methods, together with asymmetric PER, along two design axes.

2.4 Low-Loss Paths and Mode Connectivity

Mode connectivity gives empirical backing to the *monotone reference path* introduced in Section 3: the route from the old-task solution θ_A^* to the joint solution θ_{joint}^* along which the past-task loss only ever rises toward its final value, rather than first climbing above it and coming back down as the bowed descent does. For such a path to be usable, the two solutions must first be joined by a corridor of low loss at all, and this is what the mode-connectivity literature reports. Garipov et al. [6] and Draxler et al. [4] showed that independently trained minima of a deep network are joined by curved paths of near-constant low loss, and Frankle et al. [5] sharpened this to straight paths for networks that share early training. For continual learning specifically, Mirzadeh et al. [20] found that the sequentially trained and the multitask solutions often lie in the same basin, connected by a straight low-loss line. A benign, low-loss route from θ^*_A to θ^*_{joint} should therefore exist in trained networks. Where it does, the stability gap reflects the optimiser straying off that route, and an intervention succeeds by keeping it there.

3 The Stability Gap as a Trajectory Problem

This section develops the claim the thesis defends: the stability gap is not one effect but two, and both are controlled by the same lever. Learning a new task permanently moves the model, and part of the past-task loss it gives up is simply the price of that move. The stability gap is the extra loss on top, a dip below the eventual settled level followed by recovery. That excess has two sources. The first is an unavoidable geometric detour: even an ideal optimiser cannot travel straight from the old solution to the shared one, so its path bows outward and the past-task loss rises and falls as the model rounds the bend. The second is a preventable overshoot that real training adds at the task boundary, where step size, gradient noise, and momentum carry the model further off course than the geometry alone requires. The subsections below make each source precise, then show that a single design choice, how the update treats the new-task gradient, governs both and organises the methods of this thesis.

3.1 The Permanent Cost and the Reference Path

Learning a second task moves the model. In a quadratic picture with task losses L_A and L_B and Hessians H_A and H_B, the joint minimum sits at displacement $\theta^*_{\text{joint}} - \theta^*_A = (H_A + H_B)^{-1} H_B \, (\theta^*_B - \theta^*_A)$ from the old optimum, and the replay loss ends higher by approximately $\frac{1}{2} d^\top H_A \, d$ for that displacement d. This rise is the permanent stability–plasticity price. Every method pays it, and it is distinct from the stability gap.

The rise can be monotone. Consider the family of objectives $L_\lambda = L_{\text{replay}} + \lambda L_{\text{current}}$ and the path $\Theta^*(\lambda)$ traced by its minimiser as λ grows from 0 to 1. Section 4 shows, with an envelope-theorem argument, that the replay loss increases monotonically along this path under a local positive-definiteness condition. The path starts at θ^*_A, ends at θ^*_{joint}, and the replay loss climbs directly to its final level with no excursion above it. We call $\Theta^*(\lambda)$ the *reference path.* Mode connectivity gives the empirical counterpart: low-loss routes between the sequential and joint solutions exist in trained networks (Section 2.4).

The reference path defines the gap. Everything the past-task loss does *above its eventual settled level, and back*, exceeds what learning the new task requires. The stability gap is exactly this transient excess: a dip below the settled accuracy level followed by recovery. The gap metrics of Section 6.1.3 operationalise this definition, and the rest of this section explains where the excess comes from.

3.2 Level One: the Path Bows

Take the most idealised optimiser available: exact gradients and vanishing step size, pure gradient flow on the joint loss, started at θ^*_A the moment the task switches. Its path still deviates from the reference. The steepest-descent direction of the joint loss and the direction toward the joint minimum disagree whenever the loss is more steeply curved in some directions than in others, so the flow bows outward, leaves the old task's basin, crosses a region of higher replay loss, and approaches the joint minimum from outside [14]. Along this bow the replay loss rises above its final level and returns, an out-and-back *arc.* Figure 1.1 draws it as the green curve.

The arc is the structural core of the gap, for three reasons. It is deterministic: gradient noise, learning rate, and momentum play no role in it, so it survives exact replay gradients and careful step sizes. It explains the persistence results of Section 2.2: a perfect objective changes which surface is descended, but leaves the descent direction, and with it the bow, in place. And its cost scales with recovery time: the arc shows up as a dip that stays open while the iterate walks around the bow, so it dominates on networks that reconverge slowly. Fixing the arc requires changing which path the dynamics define; following the bowed path more carefully leaves it untouched.

3.3 Level Two: Overshoot of the Followed Path

Level One held the optimiser to an ideal: exact gradients, vanishing steps, no momentum. Real training meets none of these conditions, and each departure pushes the iterate beyond the path its own dynamics define, adding overshoot on top of the bow. Three mechanisms drive this overshoot, and they differ in when they act. Two are acute, set off by the task boundary. When the task switches, the objective jumps in one step from the replay loss to the joint loss, so the minimum the parameters chase teleports from θ_A^* to θ_{joint}^* while the parameters stand still. The iterate is left where the replay gradient is near zero, by convergence on Task A, and where the new-task gradient is at its largest, because Task B is unlearned [16]. This lopsided push, strongest in the first steps after the switch, is what the two acute drivers exploit. The third driver is chronic. It comes from replaying a finite buffer, is present at every step rather than triggered by the switch, and feeds the slow tail rather than the sharp drop.

Imbalance $\times$ step size (acute). With a finite learning rate, the first discrete steps ride the unopposed new-task gradient far past where the trajectory would be, a single large leap out of the old basin with a steep recovery once the replay loss kicks in. This is the mechanism behind the sharp first-step drop.

Momentum accumulation (acute). Momentum accumulates the sustained early push toward roughly $1/(1-\beta)$ times its per-step size. At the boundary the sustained direction is the provisional, old-task-degrading one, so the amplification lands exactly where it hurts.

Estimator variance (chronic). The buffer gradient is a noisy version of the true past-task gradient (Section 2.1), and it stays noisy for the whole task rather than only at the switch. Noise in the restoring force lets the iterate wander off the path and settle back slowly, so its mark is a lingering residual and a slow, jittery recovery rather than the first-step drop.

Two properties matter. First, the drivers interact: the displacement grows roughly with the size of the push, the number of steps before the restoring gradient bites, and the amplification on top, with noise jittering the whole excursion. They are mechanisms of one process, and their contributions resist clean separation into additive shares. Second, overshoot is path-agnostic: it inflates the arc of the bowed path, and it is the entire residual transient around the reference path once the arc is gone.

The two levels also assign the gap's two dimensions. The depth of the drop is set by the acute drivers; the hanging tail is the arc of Level One plus the slow, noisy recovery left by the chronic driver. This yields the falsifiable core of the thesis, tested as RQ1: removing the overshoot drivers should flatten the drop and leave the arc; placing the optimiser on the reference path should remove the arc and leave the overshoot; doing both should remove the gap while the permanent cost of Section 3.1 remains.

3.4 Two Gates

The unified update of Equation (2.1) shows that replay methods differ only in how they weight the two gradients. One asymmetry in the problem dictates how the weights should be used. The replay gradient is the only force pulling the iterate back toward the reference path, so attenuating it works against retention; the new-task gradient is the source of both the bow and the overshoot, so attenuation belongs there. Generalising the current-task weight from a scalar to an operator G gives the shared form

$$d = G\, g_{\text{cur}} + g_{\text{rep}}, \tag{3.1}$$

a *gate* on the new-task push with the restoring force at full strength. A symmetric alternative would filter the whole update through past-task curvature, braking the restoring force as hard as the harmful push. But the restoring force is exactly what pulls the model back onto the reference path, so braking both at once protects the past task only by stalling the model. The gate must therefore be one-sided: hold back the new-task push and leave the restoring force untouched.

Every method of Section 2.3 instantiates Equation (3.1), and two design axes span the family. The gate's *domain* is scalar or directional: a scalar gate attenuates the whole push, so protection is paid in plasticity, while a directional gate attenuates per direction, passing the push where the past task is flat and braking it where the past task is steep, protection nearly for free when the metric is right. The gate's *control* is feedback or feedforward: a feedback gate reads a live damage signal and reacts to harm from any direction, while a feedforward gate commits to a schedule or a precomputed filter and is only as good as its model of where harm lives.

The two methods of this thesis are the pure corners. The **adaptive λ-curriculum** is the scalar feedback gate: $\lambda(t)$ rises with the ratio of replay to new-task gradient norms, a signal that fires whichever direction damage arrives from. Because λ scales the loss itself, the gate deforms the objective into the homotopy of Section 3.1: it schedules the landscape so that the route the optimiser naturally takes is the reference path. **Asymmetric PER** is the directional feedforward gate: it filters the push through the replay Fisher, Equation (1.2), with the damping δ as its single knob. At $\delta \to \infty$ the filter opens fully and plain ER returns; at $\delta \to 0$ motion concentrates in directions the past task is flat in, while restoration keeps full strength. Table 3.1 places the replay methods on the two axes and marks where each corner sits.

Method	Gate G	Domain	Control
Vanilla ER	I	—	—
GEM / A-GEM	reweight on conflict	scalar	feedback
MEGA	loss ratio	scalar	feedback
Balancing	norm matching	scalar	feedback
Linear curric.	$\lambda(t)\,I$, clock	scalar	feedforward
Adaptive curric.	$\lambda(t)\,I$, norm ratio	scalar	feedback
Asym. PER	$\delta(F_{\text{rep}} + \delta I)^{-1}$	directional	feedforward

Table 3.1: Replay methods as gates on the new-task gradient, Equation (3.1). The two methods of this thesis occupy the pure corners: a scalar feedback gate and a directional feedforward gate.

The taxonomy yields testable hypotheses about each corner, which the research questions of Section 5 formalise and the experiments then test. The scalar feedback gate is hard to blind-side but slows the new task everywhere, so it should pay a plasticity cost that a separate accelerator must repay. The directional gate should protect the past task at little plasticity cost, but only within what its sampled curvature metric sees; damage arriving through directions the metric rates as flat passes unbraked. The two corners should also interact oppositely with momentum. The curriculum attenuates the push *at source*: with $\lambda \approx 0$ early, momentum finds nothing to accumulate, and by the time λ grows the trajectory points the right way, so acceleration helps. The preconditioner attenuates each *step* while the underlying gradient keeps its size, so accumulation across steps can rebuild the push the filter removed, and momentum should partially undo the protection. Sections 5 and 6 turn these hypotheses into the research questions and the experimental design.

4 Theory: the Monotone Reference and the $O(1/N)$ Tracking Bound

This section supplies the formal ground for the two-level account. Part A constructs the reference path of Section 3.1 and proves that the replay loss rises monotonically along it, which makes the arc of Section 3.2 avoidable in principle. Part B shows that a slow λ-ramp keeps a real SGD iterate close to this path, with a tracking error of order $1/N$ in the ramp length. The full derivation is in Appendix D.

4.1 Part A: the Reference Path Rises Monotonically

Consider the curriculum objective $L_\lambda = L_{\text{replay}} + \lambda\, L_{\text{current}}$ and the path $\Theta^*(\lambda)$ of its minimiser as λ grows from 0 to 1. The path starts at the old-task optimum ($\lambda = 0$) and ends at the joint optimum ($\lambda = 1$). Figure 4.1 shows it directly: as λ grows, the basin of L_λ morphs continuously from the old-task basin into the joint basin, and the minimiser slides along a connected valley floor between the two.

A model that follows this path exactly experiences a smooth replay loss. Appendix D shows, using the envelope theorem, that the rate of change of the replay loss along the path satisfies

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0 \qquad \text{for all } \lambda \in [0, 1] \quad (4.1)$$

under a local positive-definiteness condition made precise below. The replay loss climbs monotonically to its permanent level at the joint optimum. Together with Section 3.1, this splits the transition cleanly: the endpoint value is the fixed stability–plasticity price, and any rise *above* that value, the arc and the overshoot, is a deviation from this path.

The guarantee is local. The tracked path must remain a family of local minima, with the combined Hessian positive-definite along the branch the iterate follows. This holds automatically near $\lambda = 0$ and persists until the new task's negative curvature overpowers the old task's; beyond that point the monotonicity guarantee lapses (Appendix D). On a deep non-convex backbone such as the CIFAR ResNet the result therefore serves as motivation rather than a guarantee.

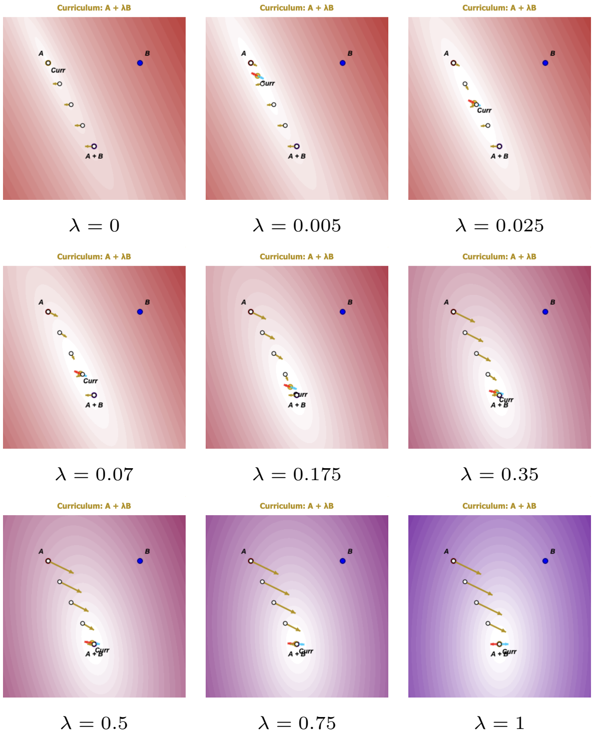

Figure 4.1: Evolution of the curriculum loss $L_\lambda = A + \lambda B$ over a shared two-dimensional parameter space, with the weight λ printed beneath each panel. A and B mark the single-task minima and $A{+}B$ the joint minimum. As λ increases, the contours morph continuously from the old-task basin around A to the joint basin around $A{+}B$, and the current minimiser (*Curr*) slides from A toward $A{+}B$: this is the reference path $\Theta^*(\lambda)$. At the minimiser the red and blue arrows (the replay and current-task gradient contributions) cancel by the first-order condition $\nabla L_{\text{replay}} + \lambda \nabla L_{\text{current}} = 0$. The golden arrows show their resultant, which vanishes at the minimiser and points down the valley away from it, so a tracking model rides the moving valley floor instead of being teleported across the landscape.

4.2 Part B: a Slow Ramp Keeps the Model on the Path

Part A describes an ideal path; a real SGD iterate $\theta(t)$ takes discrete steps and lags behind it. Analysing the gradient flow shows that the iterate behaves as if tethered to the path: its distance from the moving minimiser is proportional to how fast the valley floor moves. Stretching the transition over a ramp of N steps slows the floor down, and the maximum lag obeys

$$\|\theta(t) - \Theta^*(\lambda(t))\| = O\left(\frac{1}{N}\right). \tag{4.2}$$

4.3 The Combined Bound

Combining the monotone path (Part A) with the tracking error (Part B) bounds the highest replay loss the model experiences during the transition:

$$\max_t L_{\text{replay}}(\theta(t)) \;\leq\; L_{\text{replay}}(\theta^*_{\text{joint}}) + O\left(\frac{1}{N}\right). \tag{4.3}$$

The first term is the permanent price; the second is the transient, and it shrinks with the ramp length.

The bound reads naturally in the two-level language of Section 3. An instantaneous switch (N small) puts the optimiser on the bowed steepest-descent path of the abruptly assembled joint loss (level one) and launches it from the maximally imbalanced boundary state (level two). A slow ramp addresses both at once. The moving minimiser replaces the fixed one, so the flow path becomes the reference path and the arc disappears. And the new task enters at weight $\lambda \approx 1/N$, so the first steps carry a small push: there is little for a discrete step to leap along and little for momentum to accumulate. What the continuous-limit argument leaves out is the per-step noise of the finite buffer; the stochastic part of the overshoot survives any ramp length and needs a separate remedy, which the momentum experiments of Section 6.2.3 provide.

5 Research Questions

The two-level account of Section 3 and the bound of Section 4 make falsifiable predictions about the gap and about each gate. The experiments are organised around four questions. Each is evaluated first on rotated MNIST with a small multilayer perceptron, where the gap can be probed step by step across many conditions and seeds, and then, for the fourth question, on a deeper backbone and a stronger task shift.

RQ1 — The account. Does the stability gap separate into a deterministic arc of the descent path and overshoot of the followed path? The account predicts that the two levels respond to different interventions and are jointly responsible for the gap. Removing the overshoot drivers by construction (balanced gradients, exact replay gradients) should flatten the drop and leave the arc. Placing the optimiser on the reference path (the curriculum) should remove the arc and leave the overshoot. Applying both should remove the gap, while the permanent stability–plasticity cost remains visible elsewhere, as reduced plasticity.

RQ2 — The scalar feedback gate. Does the λ-curriculum reduce the gap by timing the new-task push, and how do its ingredients divide the work? Four sub-questions probe the design:

- **RQ2a (linear ramp).** Does gap depth fall as the ramp length N grows, consistent with the $O(1/N)$ bound?
- **RQ2b (adaptive schedule).** Does the gradient-norm feedback signal match the best fixed ramp while removing the knob N?
- **RQ2c ($\lambda_{\min}$ floor).** Does a small floor $\lambda_{\min} > 0$ recover early plasticity at a modest depth cost?
- **RQ2d (momentum).** Does momentum compose with the curriculum, closing the stochastic residual and restoring final accuracy, while leaving the depth of vanilla ER unchanged? The account predicts this composition because the curriculum attenuates the push at source, leaving momentum a clean signal to average and accelerate.

RQ3 — The directional feedforward gate. Does asymmetric PER reduce the gap as its filter strengthens, and where does a feedforward gate fail? The account predicts that depth and arc shrink monotonically as $\delta \to 0$ at little plasticity cost, and it predicts two failure modes: damage that passes through directions the sampled Fisher rates as flat, and a partial undoing of the per-step protection under momentum, because accumulation rebuilds the push the filter removes at each step.

RQ4 — Frontier and generalisation. How do the two gates compare on the stability–plasticity frontier, and do the findings carry to a deeper backbone (ResNet-18) and a stronger task shift (CIFAR-10 domain corruption)? This includes the momentum predictions of RQ2d and RQ3 as a cross of momentum settings on the deeper backbone.

6 Methods

This section describes the empirical testbed (Section 6.1.1), the metrics that summarise each run (Section 6.1.3), and the experimental blocks (Section 6.2) that answer the research questions of Section 5. Each block maps to one research question and to a prediction stated earlier.

6.1 Experimental Setup

6.1.1 Testbed

The experiments run on two testbeds of different scale. A small, fast testbed, rotated MNIST on a two-layer MLP, carries the dense mechanistic sweeps: the driver block, the curriculum sweep, the asymmetric-PER damping sweep, and the momentum cross. A larger testbed, a domain-shift CIFAR-10 on a CIFAR-adapted ResNet-18, tests whether the findings survive a deeper backbone and a stronger task shift, the regime where the local condition behind the bound of Section 4 weakens. Both testbeds share one design choice: the label space and task structure are held fixed and only the input distribution shifts, and the headline sequences use two tasks, so there is exactly one transition and exactly one instance of the stability gap.

Datasets. The small benchmark is rotated MNIST (rot-MNIST). Following Lange et al. [16], the first task T_0 presents the digits at $-90°$ and the second task T_1 applies a further $90°$ rotation. Each task uses the standard 60,000 training and 10,000 test images. The $90°$ rotation separates the two input distributions well while keeping the tasks structurally identical.

The large benchmark is a domain-shift CIFAR-10: T_0 uses clean CIFAR-10 and T_1 applies the Gaussian-noise corruption at severity 3 [10]. The ten classes are shared, so only the input distribution shifts. A three-task variant $([\text{none}, \text{Gaussian noise}, \text{shot noise}])$ and two alternative corruptions are used in the generalisation block of Section 6.2.5.

Models. The rot-MNIST sweeps use a two-layer MLP with ReLU activations; the CIFAR-10 block uses a CIFAR-adapted ResNet-18 (full specifications in Appendix A.2, Tables A.1 and A.2). The MLP is cheap to train, which matters because the sweeps span many conditions and seeds, and it is the setting where the local condition of Section 4 is most plausible. The ResNet-18 is the deeper, more realistic counterpart.

Training protocol. rot-MNIST runs in the online regime of one epoch per task, standard for stability-gap analysis: it gives a single clean pass through the transition. The ResNet-18 needs several passes to converge before the transition, so the CIFAR block uses ten epochs per task. Optimiser, learning rate, and batch size are held common across the two benchmarks; the full protocol is in Appendix A.2, Table A.3.

6.1.2 Evaluation Protocol

Throughout training, every task's test accuracy is evaluated every 10 steps. At each task transition the cadence increases to every step, so the dip is recorded at full resolution where it occurs. On rot-MNIST the per-step cadence covers the whole new task (one online epoch, ≈234 steps). On CIFAR-10 it is held for a 250-step window after the transition and then drops back to every 10 steps. A stability-gap tracker snapshots the pre-task accuracy on each past task before the transition and records the resulting (step, accuracy) series, from which it computes the scalar metrics below. A gradient tracker records, at the same cadence, the fidelity of the replay-buffer gradient against the exact past-task gradient; it is enabled for the driver block only, because each diagnostic step costs a full pass over the past-task training set.

6.1.3 Metrics

All metrics are recorded automatically and aggregated later. Each is reported as the seed mean ± sample standard deviation. Complete definitions of every logged metric are collected in Appendix B.

Stability-gap metrics. The definition of the gap in Section 3.1, the transient excess below the level the task settles at, translates directly into the two primary metrics. Let a_j^{pre} denote the baseline accuracy on past task j before the switch and $a_j(t)$ its accuracy at step t of the new task. The first metric, `gap_depth`, is the maximum instantaneous excursion,

$$\texttt{gap\_depth} = \max_j \left(a_j^{\text{pre}} - \min_t a_j(t) \right),$$

which Section 3.3 predicts to be overshoot-dominated. The second, `gap_area_end`, is the time-integral of the excursion below the settled level: with the reference point $b_j = \min\left(a_j^{\text{pre}}, \bar{a}_j\right)$, where $\bar{a}_j$ is the stable accuracy the task eventually reaches, it integrates the shortfall $\max(0, b_j - a_j(t))$ over the new task and sums across past tasks. Anchoring to the recovered level implements the split of Section 3.1: a task that steps down to a permanently lower plateau contributes its loss to the standard forgetting metrics and approximately zero to the area, while a true dip-and-recover transient registers fully. The area is where the arc of Section 3.2 and the lingering stochastic residual appear. The same min that keeps permanent forgetting out of the area has a side effect in the opposite case. When a past task recovers to or above its pre-switch baseline, the reference stays pinned at a_j^{pre}, so an early dip below baseline still accrues area even though the task ends up no worse than it started. Two methods that settle at different levels are then measured against different lines: one that recovers above baseline is scored against the baseline, while one that settles lower is scored against its own reduced plateau, so the better-recovering method can carry the larger area. A positive area therefore marks transient disruption, not a net cost, and we read it together with final accuracy rather than on its own. A third logged metric, `recovery_steps`, appears in a few places in the results; it and the gradient-fidelity diagnostics used in the driver block are defined in Appendix B.

Continual-learning metrics. To place the gap in the standard continual-learning frame, we report the continual-evaluation metrics of Lange et al. [16], computed from the task-boundary accuracy matrix logged for every run. Three carry the argument: average final accuracy (ACC), which checks that a gate buys a lower gap at preserved accuracy; worst-case retention (min-ACC, WC-ACC); and short-horizon plasticity (WP_{100}). FORG and the windowed WF_w/WP_w family are logged for completeness. All definitions are in Appendix B.

Statistical comparison. Across conditions we compare metrics with the paired Wilcoxon signed-rank test on the seed-paired difference. Seeds are paired by index, so the per-seed deltas factor out the seed-induced variance that would otherwise dominate a five-seed sample. We prefer Wilcoxon over the paired t-test because `gap_depth` and accuracy are bounded in $[0, 1]$ and typically right-skewed near zero, so the t-test's normality assumption is suspect. With five paired seeds the smallest possible two-sided p-value is 0.0625, reached exactly when all five differences share a sign; we therefore read $p = 0.0625$ as complete directional agreement and let effect sizes separate large results from marginal ones. For equivalence-style claims we report the seed-paired difference with its bootstrap 95% confidence interval.

6.2 Experiments

Each block maps to one or more research questions (Table 6.1) and tests the corresponding predictions of Sections 3 and 4. All conditions use five seeds and the protocol of Section 6.1.1 unless noted otherwise.

6.2.1 Overshoot Drivers

Aim. Remove the overshoot drivers of Section 3.3 by construction, one at a time, and observe what remains. This supplies the driver half of RQ1; the path half comes from the curriculum conditions, and the two halves meet in the analysis.

Design. Table 6.2 lists the four conditions. Balancing (D3, D4) gives g_{new} and g_{replay} equal magnitude from the first step, which switches off the boundary imbalance: an unbalanced step at θ_0^* is driven almost entirely by g_{new}, whereas a balanced step carries the replay direction at equal weight. The full-data buffer (D2, D4) replaces vanilla ER's noisy 256-sample replay gradient with the exact empirical past-task gradient, which switches off the estimator variance. The mechanics of both are in Appendix A.3.

How the conditions are read. The drivers interact (Section 3.3), so we read the conditions as interventions rather than as an additive decomposition. Depth differences between conditions measure how much of the drop each driver carries in that context. The area, and the shape of the per-step curve, carry the level-one signal: with variance switched off (D2, D4) any dip-and-recover bend that remains in the curve is deterministic, which is the arc's fingerprint.

6.2.2 The λ-Curriculum Sweep

Aim. Test the scalar feedback gate: the $O(1/N)$ prediction of Equation (4.3) (RQ2a), the adaptive schedule (RQ2b), the $\lambda_{\min}$ floor (RQ2c), and, with the full-buffer diagnostic, the claim that the curriculum removes the arc and leaves only stochastic overshoot (RQ1).

Design. All curriculum conditions run on top of standard ER (1k reservoir buffer, unbalanced gra-

Block	Section	RQ	Conditions
Overshoot drivers	6.2.1	RQ1	D1–D4
λ-curriculum sweep	6.2.2	RQ2a–c	C1–C7
Full-buffer diagnostic	6.2.2	RQ1, RQ2d	C8, C8$_M$
Momentum cross	6.2.3	RQ2d, RQ3	every rot-MNIST condition at $\mu \in \{0.0, 0.9\}$
Asymmetric-PER δ sweep	6.2.4	RQ3	$\delta \in \{1.0, 0.3, 0.1, 0.03\}$ × buffer legs, plus a solver control
CIFAR-10 generalisation	6.2.5	RQ4	vanilla, curriculum, asym. PER × {BN, GN}; momentum cross; novel corruptions

Table 6.1: Overview of the experimental blocks, the research question each answers, and the conditions in each.

Condition	Replay buffer	Gradient handling	Driver removed by construction
D1 — Vanilla ER	1k reservoir	unbalanced	none (all drivers active)
D2 — Full-data ER	60k (all past)	unbalanced	estimator variance (exact replay gradient)
D3 — Balanced ER	1k reservoir	unit-norm balanced	boundary imbalance
D4 — Full-data + balanced	60k	unit-norm balanced	imbalance and variance

Table 6.2: Driver-removal conditions. Each condition switches off one or two overshoot drivers relative to vanilla ER; what all four leave in place is the descent path itself, so the arc should survive throughout.

dients), so the schedule is the only changed variable (Table 6.3).

The adaptive schedule (C4) is the feedback variant: it sets λ to a clipped exponential moving average of the gradient-norm ratio $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$. At θ_0^* the replay gradient is near zero by stationarity, so r and λ start near 0. As the iterate moves, $\|g_{\text{replay}}\|$ grows while $\|g_{\text{new}}\|$ shrinks, so $\lambda \to 1$ at the pace the damage signal dictates, with no ramp length to choose. The floor $\lambda_{\min}$ (C5–C7) keeps the schedule from sitting at $\lambda \approx 0$ during the first steps, when the EMA of r is still close to zero, and thereby buys back early plasticity.

Full-buffer diagnostic (C8). C8 combines the shipping curriculum (C7) with the full 60k buffer of D2/D4, so the path is fixed *and* the variance driver is switched off at the source. Its role is diagnostic: if the curriculum's remaining transient at momentum 0 is stochastic overshoot, C8 should close the gap with no momentum at all, and the comparison of C8 with C8$_M$ separates momentum's two candidate roles, variance reduction (replaceable by the full buffer) and acceleration (which only momentum provides). C8 stores the entire past task and therefore breaks the limited-memory premise of continual learning; it is a probe, and C7 remains the practical configuration.

6.2.3 Momentum Cross

Every rot-MNIST condition, the driver D-series, the curriculum C-series, and the asymmetric-PER P-series, is run at momentum 0.0 and 0.9; momentum-on runs carry an M subscript (e.g. D1$_M$). The cross tests the composability prediction of Section 3.4: whether momentum helps or hurts a method's gap should depend on where the method attenuates the push. This yields the three signatures of RQ2d and RQ3, vanilla depth left unchanged, the curriculum's residual composed away, and the directional gate's per-step protection re-inflated.

6.2.4 Asymmetric-PER δ Sweep

Aim. Test the directional feedforward gate (RQ3): does the gap shrink as the filter strengthens,

Label	Condition	$\lambda(t)$ schedule	Hyperparameter	Tests
—	Vanilla ER (baseline)	$\lambda \equiv 1$ from step 1 (abrupt switch)	—	reference
C1	Linear N=50	clip(t/50, 0, 1)	$N = 50$	RQ2a
C2	Linear N=100	clip(t/100, 0, 1)	$N = 100$	RQ2a
C3	Linear N=200	clip(t/200, 0, 1)	$N = 200$	RQ2a
C4	Adaptive	clip(EMA$_\alpha$(r), 0, 1)	$\alpha = 0.05$	RQ2b
C5	Adaptive + $\lambda_{\min}$=0.05	max($\lambda_{\min}$, clip(EMA$_\alpha$(r), 0, 1))	$\alpha = 0.05$, $\lambda_{\min}$=0.05	RQ2c
C6	Adaptive + $\lambda_{\min}$=0.10	as C5	$\lambda_{\min}$=0.10	RQ2c
C7	Adaptive + $\lambda_{\min}$=0.20	as C5	$\lambda_{\min}$=0.20	RQ2c
C8	C7 + full 60k buffer	as C7, exact replay gradient	diagnostic	RQ1, RQ2d

Table 6.3: λ-curriculum conditions, all applied on top of standard ER. EMA$_\alpha$ **is an exponential moving average with smoothing α, and $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ is the gradient-norm ratio. Each condition is run at momentum 0.0 and 0.9.**

at what plasticity cost, and do the two predicted feedforward failure modes appear?

Method. Asymmetric PER implements Equation (1.2): it filters the new-task push through the replay Fisher F_{rep}, the model's own estimate of which directions the past task depends on [12]. Normalising by δ keeps flat directions at unit gain: along an eigendirection of curvature ρ the gain is $\delta/(\rho+\delta)$, so directions the past task is flat in pass through whole and steep ones are shrunk toward zero. The damping δ is the single knob: at $\delta \to \infty$ every direction passes and plain ER returns, while as $\delta \to 0$ the push concentrates in the replay-flat directions with the restoring gradient g_{rep} at full strength throughout. The push is applied matrix-free by conjugate gradients over Fisher-vector products; solver details are in Appendix A.4.

Design. The sweep crosses the damping $\delta \in \{1.0, 0.3, 0.1, 0.03\}$ with two buffer legs, at both momentum settings (Section 6.2.3). The *standard* leg uses the 1k reservoir buffer and matches the practical regime of the curriculum sweep. The *full-buffer* leg makes the replay gradient exact, switching off the variance driver as in D2/D4, so the remaining area measures the deterministic arc and the sweep reads out directly whether the directional gate shrinks the arc itself. A solver control (Appendix A.4) confirms the results are not an artefact of an under-converged solve.

6.2.5 Generalisation: CIFAR-10

Aim. Test whether both gates and the momentum predictions carry to a deeper backbone and a stronger task shift (RQ4). The quantitative $O(1/N)$ scaling is a local result and is checked only qualitatively here.

Protocol. All CIFAR conditions use the ResNet-18 with the CIFAR-10 protocol of Table A.3 (ten epochs per task, buffer 5,000) and the evaluation scheme of Section 6.1.2. The headline conditions run at momentum 0.9, the standard setting for this backbone. Each method is crossed with BatchNorm and GroupNorm, because the two normalisations respond differently at the switch: the input shift disrupts BatchNorm's running statistics, which must re-adapt during the first steps of the new task, while GroupNorm uses no cross-batch statistics, so its transient reflects optimisation dynamics alone.

Headline block. Table 6.4 (top) compares the three methods at both normalisations. The curriculum ships as the adaptive schedule at $\lambda_{\min} = 0.20$, the configuration selected on rot-MNIST; asymmetric PER ships at $\delta = 0.1$, a strong rot-MNIST setting on both axes that stays clear of the blind-spot regime of the sweep. Both methods carry their rot-MNIST configurations over without a CIFAR-specific sweep, so they meet the deeper backbone on equal footing.

Label	Method	Norm	μ
G1	Vanilla ER	BatchNorm	0.9
G3	Adaptive $\lambda_{\min}$=0.20	BatchNorm	0.9
G4	Vanilla ER	GroupNorm	0.9
G6	Adaptive $\lambda_{\min}$=0.20	GroupNorm	0.9
G7	Asym. PER δ=0.1	BatchNorm	0.9
G8	Asym. PER δ=0.1	GroupNorm	0.9
G9	Vanilla ER	GroupNorm	0.0
G10	Adaptive $\lambda_{\min}$=0.20	GroupNorm	0.0
G11	Asym. PER δ=0.1	GroupNorm	0.0

Table 6.4: CIFAR-10 conditions (two-task Gaussian-noise shift, buffer 5,000, five seeds). Top: the headline block at momentum 0.9. Bottom: the momentum cross, which pairs momentum-0 runs of all three methods against their momentum-0.9 counterparts at GroupNorm.

Momentum cross. Table 6.4 (bottom) adds momentum-0 runs of all three methods at GroupNorm, whose transient is free of normalisation-statistics effects. Together with the momentum-0.9 rows, this repeats the three-way composability test of Section 6.2.3 on the deep backbone: depth unchanged for vanilla, composition for the curriculum, re-inflation for asymmetric PER. Momentum also drives convergence on this backbone, so accuracy is compared within a momentum setting rather than across.

Generalisation block. A further block tests whether the curriculum's result depends on the particular corruption or on having a single transition. It pairs the shipping curriculum against a vanilla-ER control on a two-task shot-noise shift, a two-task contrast shift, and a three-task sequence [none, Gaussian noise, shot noise], each at both normalisations over three seeds (Table 6.5). The paired control keeps the comparison interpretable: the question is whether the gate still beats vanilla on data it was never tuned on.

Shift	Tasks	Conditions
Shot noise	2	ship vs. vanilla
Contrast	2	ship vs. vanilla
Gaussian noise + shot noise	3	ship vs. vanilla

Table 6.5: CIFAR-10 generalisation block. Each shift runs the shipping curriculum (adaptive $\lambda_{\min}$=0.20) and a paired vanilla-ER control, at both normalisations, three seeds.

7 Results

This section reports the experimental blocks of Section 6 in the order of the research questions: the two-level account (Section 7.1, RQ1), the scalar feedback gate (Section 7.2, RQ2), the directional feedforward gate (Section 7.3, RQ3), the stability–plasticity frontier the two gates trace (Section 7.4, RQ4a), and the CIFAR-10 generalisation of both legs (Section 7.5, RQ4b).

Reading the tables. Depth is reported in percentage points (pp); a depth of 19.5 means the worst past-task evaluation falls 19.5 points below its pre-switch baseline. Section 3.3 predicts depth to be overshoot-dominated, so it reads the perceived severity of the drop. The end-referenced area `gap_area_end` (in accuracy·steps) integrates the excursion below the level each task settles at, so a permanent accommodation contributes to the standard forgetting metrics and near-zero to the area; Section 3.2 predicts the deterministic arc and the lingering stochastic residual to live here. Because each condition's area is taken against its own settled level (Section 6.1.3), area is directly comparable only between conditions that recover to the same level: a condition that retains a past task better is scored against a higher line and can register a *larger* area than one that settles lower, so we read cross-condition area differences together with final accuracy and the per-step curves rather than on their own. Values are mean $\pm$ standard deviation over five seeds (three for the CIFAR generalisation block). We follow the statistical convention of Section 6.1.3: with five paired seeds the two-sided Wilcoxon test floors at $p = 0.0625$ when all five differences share a sign, so we quote a p-value only where the seeds disagree and otherwise report complete directional agreement, letting effect sizes separate large results from marginal ones. The complete metric set for every condition is in Appendix C.

7.1 The Two Levels, Separated (RQ1)

RQ1 asks whether the gap separates into a deterministic arc of the descent path and overshoot of whatever path is followed, such that removing the overshoot drivers and fixing the path are separately insufficient but jointly sufficient. The test crosses the two interventions at momentum 0: the driver-removal conditions D1–D4 fix nothing about the path, and the curriculum conditions C7/C8 fix the path while leaving or removing the variance driver. Table 7.1 is the 2×2 this produces.

	Drivers present	Drivers removed
Bowed path (no gate)	D1: 19.5 / 6.3 / 0.873	D4: 1.9 / 3.2 / 0.869
Valley-floor (curriculum)	C7: 13.9 / 2.7 / 0.837	C8: 0.2 / 0.05 / 0.818

Table 7.1: The RQ1 2×2 on rot-MNIST at momentum 0, each cell reporting `gap_depth` (pp) / `gap_area_end` / ACC. Rows cross the path (bowed steepest descent vs. the valley-floor path the curriculum follows); columns cross the overshoot drivers (present vs. removed by balancing and the exact full-buffer replay gradient). Removing the drivers alone (D4) leaves a deterministic arc; fixing the path alone (C7) leaves the overshoot; doing both (C8) closes the gap, while the permanent stability–plasticity cost stays visible as the lowest ACC in the table.

Removing the drivers leaves the arc. Table 7.2 follows the driver-removal path. Unit-norm balancing (D3) removes the boundary imbalance and takes 13.8 pp off the depth ($19.5 \to 5.7$), by far the largest single drop; swapping the reservoir for the exact past-task gradient (D4) removes the estimator variance and takes off a further 3.8 pp ($5.7 \to 1.9$). The gradient-fidelity diagnostics confirm the driver each condition switches off: the finite-buffer conditions D1 and D3 reach only $\bar{\cos}(g_{\text{replay}}, g_{\text{true}}) \approx 0.71$ and 0.66 with per-step minima near zero, whereas the full-data conditions D2 and D4 sit at 1.000 by construction (Appendix C.3). What no driver removal touches is the descent path, and the area column shows this: with the depth spike gone at D4 (only 1.9 pp) a clear bend remains, `gap_area_end` = 3.2. Because the full-data conditions carry zero gradient noise, that area cannot be stochastic residual; it is a deterministic bend. It is the arc of Section 3.2, and Figure 7.1 shows it directly: the exact-gradient curves (D2, D4) trace a smooth dip-and-recover with no jitter, the fingerprint the account attaches to a bowed path rather than a spike.

Condition	`gap_depth` (pp)	`gap_area_end`	ACC
D1: Vanilla ER	19.5 ± 4.1	6.3 ± 0.5	0.873 ± 0.005
D2: Full-data ER	18.3 ± 2.5	8.1 ± 0.7	0.890 ± 0.009
D3: Balanced ER	5.7 ± 1.9	1.4 ± 0.4	0.849 ± 0.005
D4: Full-data balanced	1.9 ± 1.3	3.2 ± 0.5	0.869 ± 0.002

Table 7.2: Driver-removal conditions on rot-MNIST (momentum 0). Balancing (D3) removes most of the depth; the exact replay gradient (D4) removes the rest. Neither removes the bend: D4 has a depth of only 1.9 pp yet a deterministic arc of area 3.2.

Fixing the path leaves the overshoot. The other row of the 2×2 holds the drivers in place and fixes the path with the curriculum. At momentum 0 the shipping curriculum (C7) still shows 13.9 pp of depth (Table 7.1): with the arc removed, essentially all of it is overshoot. That this residual is the estimator variance rather than anything structural is settled by C8, which adds the exact full buffer to C7 so that the path is fixed *and* the variance driver is switched off at the source. The gap then collapses to 0.2 ± 0.3 pp of depth and 0.05 of area, the smallest of any condition in the study, the C7→C8 area drop ($2.7 \to 0.05$) holding across all five seeds. Overshoot is real and untouched by the path fix; the arc is real and untouched by the driver removal; only doing both closes the gap.

The permanent cost stays on the ledger. Closing the transient does not make the transition free. C8's final accuracy is 0.818, the lowest in Table 7.1 and 0.055 below vanilla ER's 0.873, because holding the new-task push down at momentum 0 slows the new task without an accelerator to repay it. This is the permanent stability–plasticity cost the account predicts to remain visible elsewhere on the ledger once the transient is removed. Section 7.2 shows momentum paying that debt back.

A second, independent line of evidence that the arc is deterministic geometry comes from the directional gate and is reported in full in Section 7.3: on its full-buffer legs, where the variance driver is off and the area *is* the arc, the area shrinks monotonically as the filter strengthens ($2.3 \to 1.0 \to 0.4$ for

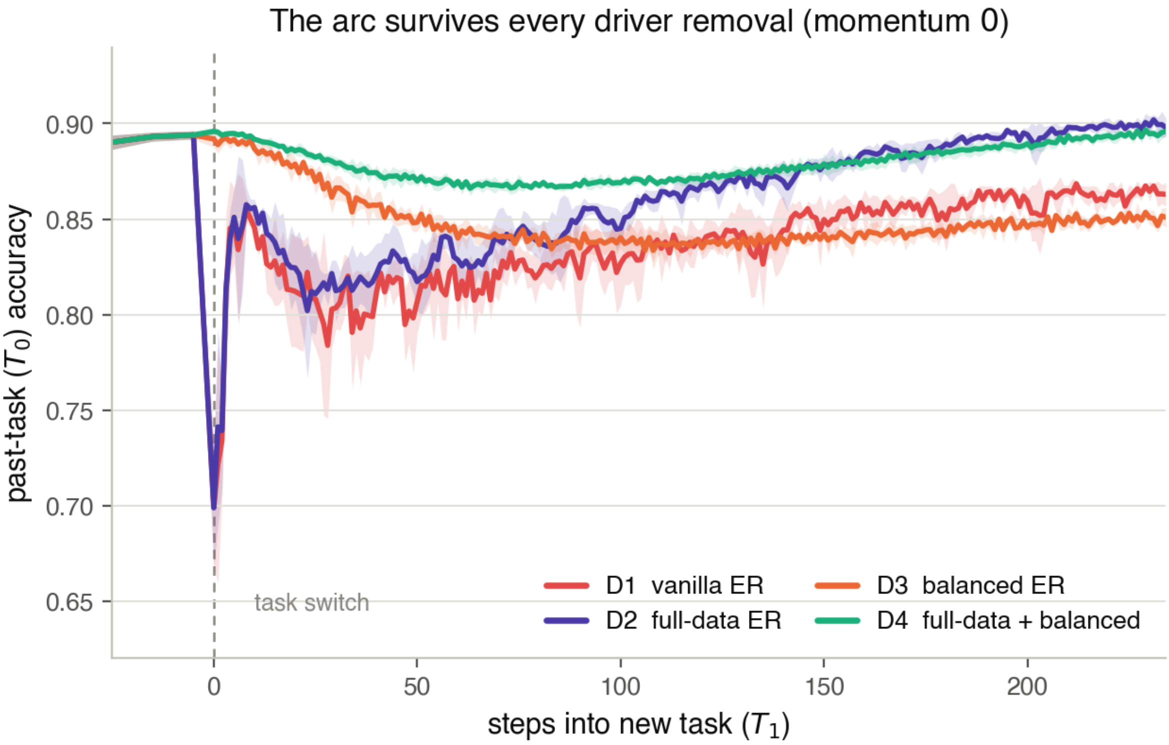

Figure 7.1: Per-step past-task (T_0) accuracy across the $T_0 \to T_1$ transition for the four driver-removal conditions at momentum 0; solid lines are the seed mean, bands the standard deviation, and $x{=}0$ marks the task switch. Two readings are encoded: *depth* of the boundary spike falls when the imbalance driver is switched off by balancing (deep for the unbalanced D1, D2; shallow for the balanced D3, D4), while *jitter* vanishes when the variance driver is switched off by the exact replay gradient (the full-data conditions D2, D4 are smooth, the finite-buffer D1, D3 noisy). What no removal touches is the smooth dip-and-recover *bend* beneath the spike, cleanest in the exact-gradient D4 where no jitter can hide it: that bend is the deterministic arc, and it responds to no amount of optimiser hygiene, only to a change of path.

$\delta = 1.0 \to 0.3 \to 0.1$). The arc responds to path-level intervention and to nothing else, whether the intervention times the push (the curriculum) or filters its direction (the gate).

7.2 The Scalar Feedback Gate (RQ2)

Table 7.3 reports the curriculum sweep at both momentum settings against the vanilla-ER baseline. The momentum-off leg tests the schedule in isolation; the momentum-on leg is the shipping regime.

Lengthening the curriculum ramp strictly reduces gap depth (RQ2a, RQ2b, RQ2c). At momentum 0, gap depth falls monotonically as the linear ramp lengthens, exactly as the $O(1/N)$ bound of Equation (4.3) predicts (RQ2a; Table 7.3), though successive ramp lengths lie within the seed spread and the reduction is bought at a real accuracy cost: by $N = 200$ ACC has fallen from 0.873 to 0.828, a drop of 0.045, and the tail has stretched, `recovery_steps` growing from 36 at $N = 50$ to 133 at $N = 200$ (Appendix C). The adaptive schedule (C4) reaches 12.8 pp, below the best linear ramp, without a ramp length to set (RQ2b; C3→C4 depth agrees in direction but stays within noise, $p = 0.44$). Raising the floor $\lambda_{\min}$ from 0 to 0.20 (C7) shifts the schedule in the predicted Pareto direction only weakly at momentum 0 — a slight accuracy gain for a comparably small depth increase — because early T_1 progress is then set by the large deterministic boundary gradient the small floor cannot redirect (RQ2c). The momentum-0 curves of Figure 7.2 (dashed) show the trade directly for the shipping schedule: the curriculum reaches a shallower past-task dip than vanilla (a) but pays for it in slower early T_1 learning (b); the linear-ramp and floor sweeps behind the numbers above are tabulated in Appendix C.

Momentum pays the plasticity debt, and closes the gap (RQ2d). The momentum-on leg of Table 7.3 is where the scalar gate ships, and it acts through momentum's two roles at once. Adding momentum to the adaptive schedule drops depth to 2.5 pp (C4→C4$_M$) and to 3.7 pp at the $\lambda_{\min} = 0.20$ floor (C7$_M$), collapses the area to near zero, and restores accuracy to vanilla ER's level (Table 7.3); every one of these contrasts holds across all five seeds. The depth and area collapse is variance reduction: momentum averages the noisy finite-buffer replay gradient across accumulated velocity, damping the chronic stochastic driver that feeds the residual dip and tail once the schedule has already tempered the acute boundary push at source. The accuracy restoration is the second, separate role — acceleration, driving the new task hard enough to repay the plasticity the schedule borrows. The two are confirmed to be distinct below: the full buffer reproduces the first without the second. The floor that was inert at momentum 0 becomes a clean Pareto knob once momentum is active: raising it from 0 to 0.20 (C4$_M$ →C7$_M$) trades 1.2 pp of depth for 0.014 of final accuracy and a gain in short-horizon plasticity (WP$_{100}$; Appendix C), because a higher floor lets T_1 train from the first post-switch step, the plasticity momentum buys back (Figure 7.2(b)). At C7$_M$ the past task no longer dips and recovers but settles at a marginally lower plateau (area ≈ 0), so the residual 3.7 pp of depth is the permanent accommodation to the new task, not a transient. This configuration ships to CIFAR.

That momentum here closes a gap it left untouched under vanilla ER is the composability signature of Section 6.2.3, and it separates momentum's two roles cleanly. Table 7.4 crosses the curriculum with the full buffer at both momentum settings. C8 (full buffer, no momentum) already has essentially no gap (0.2 pp), so the residual the curriculum leaves at momentum 0 was estimator variance, closable by source variance reduction rather than by acceleration; momentum's gap-closing role is interchangeable with the full buffer. What only momentum provides is plasticity: C8→C8$_M$ lifts ACC from 0.818 to 0.955 (the highest in the benchmark) while re-introducing a small 2.0 pp gap, the irreducible transient price of driving the new task hard even at zero noise.* The headline is that *the curriculum needs momentum for accuracy, not for*

*C8$_M$'s tail (area 0.378) exceeds C7$_M$'s (0.035), which the reference definition of Section 6.1.3 accounts for: C7$_M$'s residual is a permanent step to a lower plateau and is excluded from the area, whereas C8$_M$ recovers to the highest accuracy in the benchmark, so its 2.0 pp dip is a genuine transient the area counts in full. C8/C8$_M$ are diagnostic probes: the full 60k buffer defeats the limited-memory premise, and the practical scalar-gate configuration remains C7$_M$ (3.7 pp / 0.035 / 0.931) at 1/60th the memory.

Condition	Momentum 0.0 `gap_depth` (pp)	`gap_area_end`	ACC	Momentum 0.9 `gap_depth` (pp)	`gap_area_end`	ACC
Vanilla ER (baseline)	19.5 ± 4.1	6.3 ± 0.5	0.873 ± 0.005	15.9 ± 2.6	1.0 ± 0.2	0.928 ± 0.003
C1: Linear N=50	17.5 ± 1.7	4.9 ± 0.5	0.851 ± 0.023	6.4 ± 0.5	0.2 ± 0.1	0.932 ± 0.003
C2: Linear N=100	16.2 ± 4.3	3.6 ± 0.3	0.844 ± 0.025	6.4 ± 0.6	0.2 ± 0.1	0.932 ± 0.003
C3: Linear N=200	14.3 ± 4.6	1.8 ± 0.6	0.828 ± 0.025	6.2 ± 0.6	0.1 ± 0.0	0.925 ± 0.007
C4: Adaptive	12.8 ± 2.9	2.2 ± 0.5	0.833 ± 0.025	2.5 ± 0.5	0.3 ± 0.1	0.917 ± 0.001
C5: Adaptive $\lambda_{\min}$=0.05	13.2 ± 3.2	2.2 ± 0.5	0.835 ± 0.025	2.6 ± 0.5	0.1 ± 0.0	0.922 ± 0.001
C6: Adaptive $\lambda_{\min}$=0.10	13.9 ± 3.3	2.4 ± 0.5	0.836 ± 0.026	3.0 ± 0.4	0.0 ± 0.0	0.927 ± 0.002
C7: Adaptive $\lambda_{\min}$=0.20	13.9 ± 3.4	2.7 ± 0.6	0.837 ± 0.031	3.7 ± 0.4	0.0 ± 0.0	0.931 ± 0.002

Table 7.3: λ-curriculum sweep on rot-MNIST against vanilla ER, at momentum 0.0 and 0.9; five seeds. The scalar feedback gate reaches a few points of depth at matched accuracy only in the momentum-on regime it ships in.

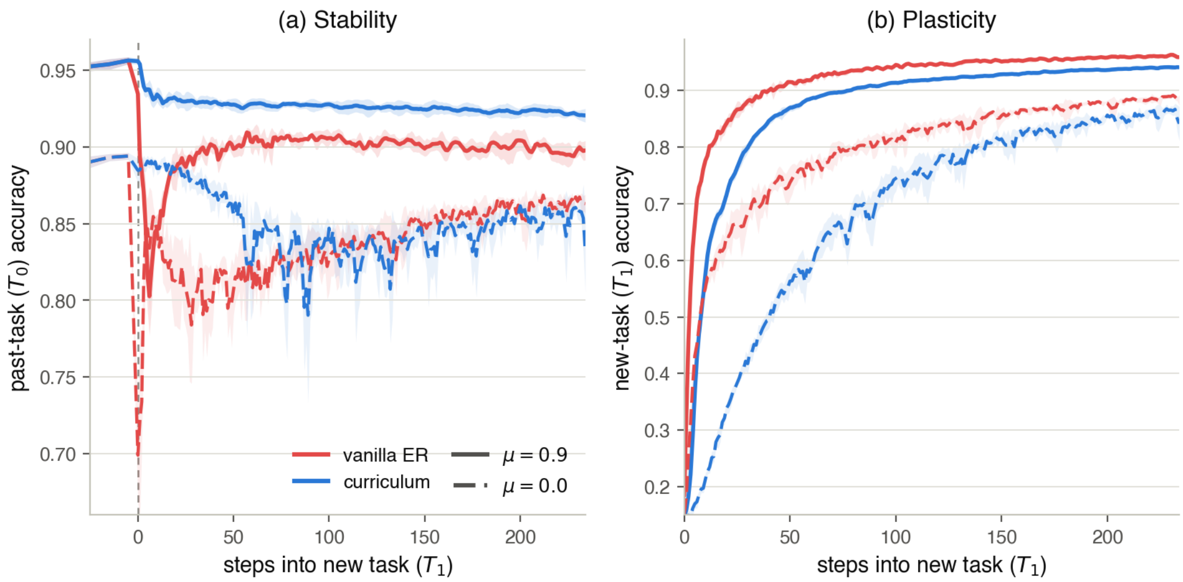

Figure 7.2: The scalar feedback gate and momentum's two roles: per-step accuracy across the rot-MNIST switch, seed mean with ±std bands, for vanilla ER (red) and the shipping adaptive curriculum $\lambda_{\min}$=0.20 (blue), each at momentum 0.9 (solid) and 0.0 (dashed). (a) Stability (T_0): only curriculum-with-momentum (solid blue) holds the past task near its plateau through the switch; momentum barely dents vanilla's spike (solid red still dives), and both methods open a deep, jittery gap without it (dashed). Momentum closes the gap where the gate has already tempered the push at source, not otherwise. (b) Plasticity (T_1): at momentum 0 the curriculum slows early new-task learning (dashed blue is the slowest to rise — the plasticity debt of holding the push down), but momentum pays it back, lifting the solid-blue learning curve back toward the vanilla pace.

the gap: at momentum 0, with the full buffer standing in for momentum's averaging, the gate is already gap-free; momentum's own contribution to the shipping configuration is the plasticity it buys back.

	C7 (1k, μ=0)	C7$_M$ (1k, μ=0.9)	C8 (full, μ=0)	C8$_M$ (full, μ=0.9)
`gap_depth` (pp)	13.9	3.7	**0.2**	2.0
`gap_area_end`	2.7	0.03	**0.05**	0.38
ACC	0.837	0.931	0.818	**0.955**

Table 7.4: Curriculum $\times$ full buffer $\times$ momentum on rot-MNIST, isolating momentum's two roles. Source variance reduction alone (C8) closes the gap; acceleration alone (the μ column) restores accuracy. The residual at C7 (μ=0) is variance, not structure.

For completeness the momentum cross also confirms the vanilla-ER null: momentum shifts vanilla depth only from 19.5 to 15.9 pp, a change smaller than the seed spread with seeds split on its sign ($p = 0.44$), because the deterministic unopposed first step carries no accumulated velocity to average. Momentum does cut vanilla's tail (area 6.3 $\to$ 1.0, all seeds). The correct statement is therefore that momentum leaves vanilla's *depth* intact while smoothing its recovery, not that it leaves the gap intact.

7.3 The Directional Feedforward Gate (RQ3)

RQ3 asks whether asymmetric PER shrinks the gap as its filter strengthens ($\delta \to 0$), at what plasticity cost, and where a feedforward gate fails. Table 7.5 reports the δ sweep on both buffer legs at momentum 0.

Monotone, and free. On the standard buffer, depth falls from 9.0 to 4.8 pp as δ goes from 1.0 to 0.03 while ACC *rises* from 0.869 to 0.880 and WP_{100} from 0.733 to 0.741; every one of these three trends holds across all five seeds, and the shrinking dips of Figure 7.3(a) show it step by step. The plasticity cost the scalar gate pays (roughly 4 pp of ACC at momentum 0, Table 7.3) never materialises here, because the filter blocks the push only in the directions the past task is steep in and lets it through everywhere else. This is the scalar gate's weakness inverted, and it is what makes the two gates complements rather than a strong and a weak version of the same thing.

The gate reshapes the path. The full-buffer leg isolates the arc: with the variance driver off, the area is the deterministic bend, and it shrinks with the filter, 2.3 $\to$ 1.0 $\to$ 0.4 for $\delta = 1.0 \to 0.3 \to 0.1$ (the 1.0 $\to$ 0.3 area drop holds across all seeds), the smooth dip pulling back toward the plateau as δ falls in Figure 7.3(b). This is the independent level-1 evidence promised in Section 7.1: the directional gate bends the followed path back toward the reference, exactly as the curriculum does by another route. The strongest clean setting, $\delta = 0.3$ full buffer, reaches 1.5 pp of depth, undercutting the driver-removal floor D4 (1.9 pp) at the best momentum-0 accuracy in the benchmark (0.900 vs. 0.869), and without disturbing the step-size adaptation that unit-norm balancing changes.

At momentum 0 the directional gate dominates the scalar gate. Head to head at momentum 0, the strongest standard-buffer directional setting (δ=0.03: 4.8 pp / 0.880) beats the shipping curriculum (C7: 13.9 pp / 0.837) on both axes at once, lower in depth and higher in accuracy. Part of this is that the curvature filter also damps how far a noisy gradient moves the past-task loss, so it suppresses the *expression* of variance per step where the scalar gate has no such side effect and needs momentum or a larger buffer for its residual (Section 7.2). The scalar gate's case is not the momentum-0 corner; it is momentum composability and blind-spot robustness, which the next two results show the directional gate lacking.

The cliffs: a feedforward blind spot. The one place the sweep breaks monotonicity is the strongest full-buffer setting, $\delta = 0.03$, where mean depth jumps to 9.5 pp with a std of 8.2: in 3 of 5 seeds the past-task curve, having held flat for two hundred steps, slides off a late cliff (`recovery_steps` ≈ 217; min-ACC std 0.094), the red per-seed traces in Figure 7.3(b). Re-running the same setting with the conjugate-gradient solver taken from its 25-iteration baseline to 60 iterations reproduces the cliffs at the same seeds and steps and at a comparable mean depth: the 60-iteration diagnostic run gives 11.5 pp, a separate confirma-

	Standard (1k) buffer				Full (60k) buffer			
δ	depth (pp)	area	ACC	WP_{100}	depth (pp)	area	ACC	min-ACC
1.0	9.0 ± 0.6	3.3	0.869	0.733	4.0 ± 1.0	2.3	0.898	0.841
0.3	7.4 ± 1.1	2.3	0.873	0.736	$\mathbf{1.5 \pm 1.1}$	1.0	0.900	0.867
0.1	5.9 ± 1.5	1.6	0.876	0.739	2.2 ± 1.6	0.4	0.901	0.860
0.03	4.8 ± 1.9	1.0	0.880	0.741	9.5 ± 8.2	0.4	0.903	0.787

Table 7.5: Asymmetric-PER δ sweep on rot-MNIST at momentum 0, five seeds. Depth and area fall as $\delta \to 0$ while ACC and WP_{100} *rise*: the directional gate protects the past task at no plasticity cost. On the full buffer, where the area is the deterministic arc, the arc shrinks with the filter, and the strongest clean setting (δ=0.3) undercuts the driver-removal floor (D4: 1.9 pp). The far cell is the exception: at δ=0.03 full buffer the mean depth jumps to 9.5 pp with a large std, the cliffs of the text.

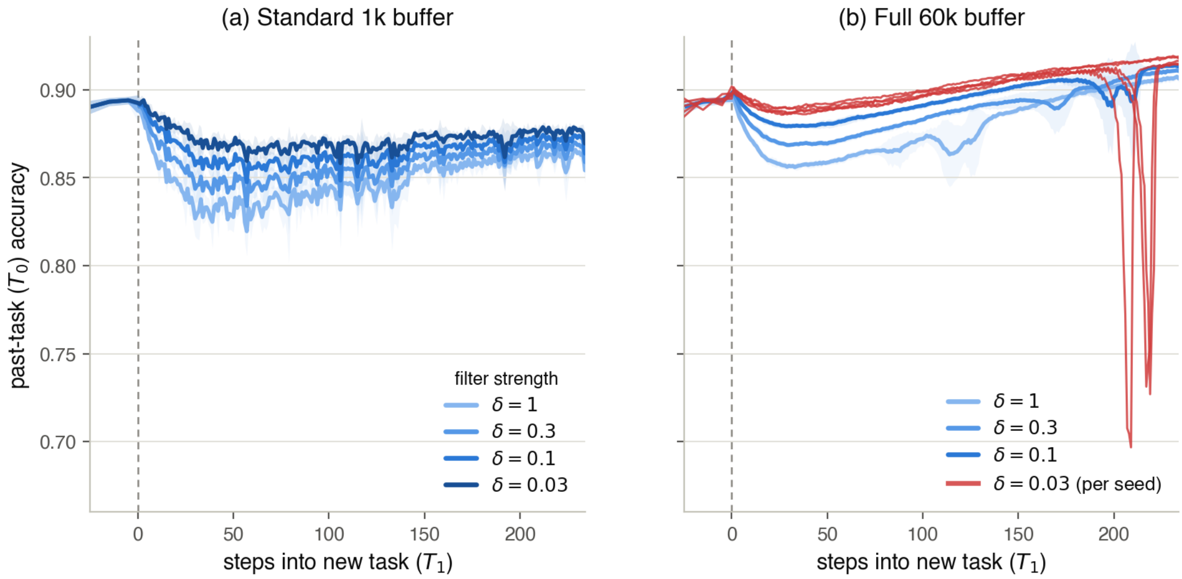

Figure 7.3: The directional feedforward gate: per-step past-task (T_0) accuracy across the rot-MNIST switch at momentum 0, sweeping the damping δ (weaker filter light $\to$ stronger filter dark). (a) Standard 1k buffer: the boundary dip shrinks monotonically as $\delta \to 0$ — and accuracy rises with it (Table 7.5) — so the protection is bought at no plasticity cost. (b) Full 60k buffer (variance driver off, so the residual excursion *is* the deterministic arc): the arc shrinks with the filter for $\delta \geq 0.1$, but at the strongest setting δ=0.03 the feedforward gate hits its curvature blind spot — shown per seed (critical red), 3 of 5 seeds hold flat for $\sim$200 steps and then slide off a late cliff, the failure a feedback signal cannot be blind-sided into.

tion value alongside the 9.5 pp of the 25-iteration baseline in Table 7.5, not a re-quote of it, so the cliffs are not an artefact of an under-converged solve. They are the feedforward failure the taxonomy predicts: new-task pressure escapes through directions the *sampled* Fisher rates as flat but the true past-task loss punishes further out. A gate that brakes only where its model of the past task says to brake is blind wherever that model is wrong, and the model is estimated on finite data under a saturating loss.[†] The cliffs are the empirical case for a feedback signal, which reads the damage whatever direction it arrives from and so cannot be blindsided.

Momentum re-inflates the per-step filter. Momentum undoes the directional gate's protection, the opposite of what it does for the scalar gate. On the full buffer, momentum lifts depth from 2.2 to 7.6 pp at δ=0.1 and from 1.5 to 9.1 pp at δ=0.3 (both across all five seeds), while accuracy jumps to $\approx$ 0.965. The filter attenuates the step, not the source: the driving magnitude of the new-task gradient is untouched, so velocity accumulated across steps rebuilds displacement in the very directions the filter shrinks one step at a time. Under momentum, then, asymmetric PER becomes the highest-accuracy replay method in the study at a depth ($\approx$ 8 pp) between the scalar gate's (3.7 pp, $C7_M$) and vanilla ER's (15.9 pp): neither gate dominates once momentum enters.

7.4 The Frontier (RQ4a)

The two gates do not order; they trace a stability–plasticity frontier. Figure 7.4 plots every rot-MNIST gate condition in the accuracy–depth plane, the λ-path (scalar feedback) against the δ-path (directional feedforward), anchored by the no-gate baseline (D1), the driver-removal floor (D4), and the diagnostic floor (C8).

Two regimes read off the figure. At momentum 0 (left panel) the directional gate dominates: its points lie above and to the right of the scalar gate's, lower in depth and higher in accuracy, with the driver floor D4 reached and undercut by the full-buffer δ-path and the cliff point the lone exception. Under momentum (right panel) the picture inverts into a genuine trade. The scalar gate reaches the lowest depths (2.5–3.7 pp) because momentum composes with its source attenuation, but caps out near vanilla accuracy; the directional gate cannot go as low in depth (momentum re-inflates it to $\approx$ 8 pp) but reaches the highest accuracies ($\approx$ 0.965), because its per-step filter never slowed the new task at source. A practitioner evaluated on worst-case retention during the transition, who can absorb the accuracy, takes the scalar gate; one who needs peak accuracy and can tolerate a modest transient takes the directional gate. The result of RQ4a is the frontier itself.

7.5 Generalisation to CIFAR-10 (RQ4b)

The CIFAR-10 block tests whether both gates carry to a ResNet-18, a clean $\rightarrow$ corruption shift, and the two normalisations that govern the post-switch recovery. Table 7.6 reports the headline block at momentum 0.9.

Both gates carry, and the frontier does not survive the backbone. The rot-MNIST result carries over and strengthens for both gates. Under batch norm the curriculum cuts depth from 11.8 to 4.8 pp and asymmetric PER to 5.9 pp; under group norm, where vanilla ER suffers a catastrophic 41.2 pp drop, the curriculum cuts it to 5.2 pp and asymmetric PER to 5.6 pp. Every gate-vs-vanilla depth and area contrast holds across all five seeds, at final accuracy within about 0.01 of vanilla either way (the curriculum's ACC contrasts split the seeds, as matched accuracy should; asymmetric PER runs about 0.01 below vanilla under batch norm and above it under group norm) and with worst-case retention sharply improved, min-ACC rising from 0.31 to 0.67 under group norm. The rot-MNIST ordering does not survive the deeper backbone: the directional gate, which was momentum's weakest regime on the MLP, here matches the scalar gate and under group norm slightly exceeds its accuracy (0.741 vs. 0.731). On this backbone the transient

[†] A finer reading—that softmax saturation on replay data drives the local Fisher toward zero while a cliff lies ahead, and that the 1k buffer's sampling noise incidentally dithers such escapes away, which is why the cliffs appear on the full-buffer leg—is consistent with the data but not established here; we report the failure at the blind-spot level the solver control supports.

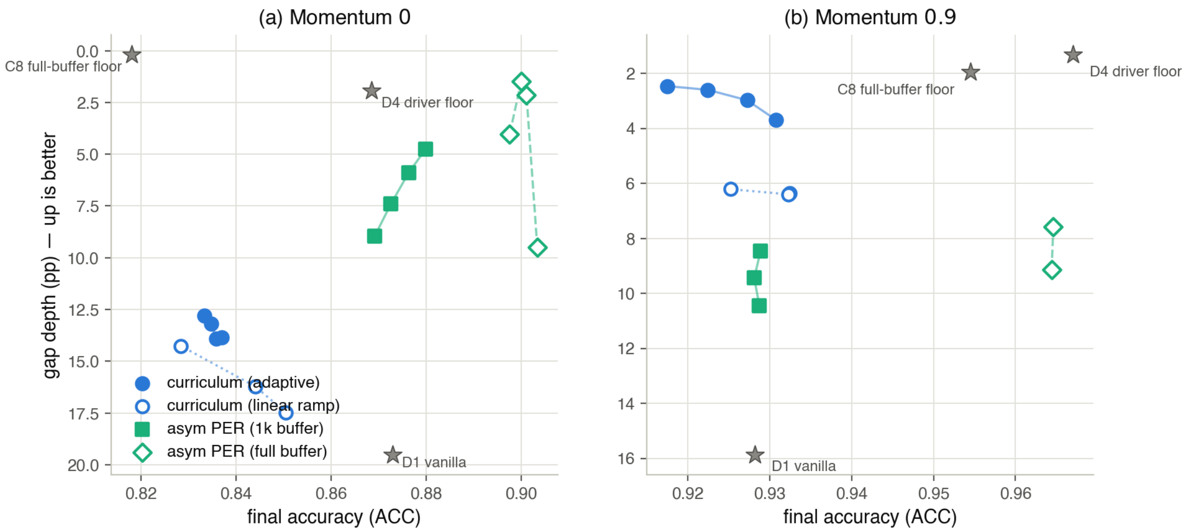

Figure 7.4: The stability–plasticity frontier on rot-MNIST: final accuracy (right is better) against gap depth (up is better, axis inverted). The λ-path (blue) is the curriculum, adaptive+$\lambda_{\min}$ as filled circles and linear ramps as open circles; the δ-path (aqua) is asymmetric PER, the 1k buffer as filled squares and the full buffer as open diamonds, each sweeping $\delta = 1.0 \to 0.03$; grey stars are the anchors (no-gate D1, driver floor D4, full-buffer floor C8). (a) Momentum 0: the δ-path dominates, sitting up and to the right of the λ-path (lower depth *and* higher accuracy) and reaching the D4 floor; the lone δ=0.03 full-buffer diamond falls away as the blind-spot cliffs open. (b) Momentum 0.9: neither dominates. The λ-path reaches the lowest depth (top-left), the δ-path the highest accuracy (right); the frontier, not a winner, is the result.

Norm	Method	gap_depth (pp)	gap_area_end	ACC	min-ACC
BatchNorm	G1: Vanilla ER	11.8 ± 2.5	17.5 ± 4.6	0.723 ± 0.008	0.644 ± 0.023
	G3: Curriculum ($\lambda_{\min}$=0.20)	$\mathbf{4.8 \pm 0.7}$	$\mathbf{3.6 \pm 2.2}$	0.722 ± 0.011	0.713 ± 0.017
	G7: Asym. PER (δ=0.1)	5.9 ± 4.3	3.3 ± 3.0	0.715 ± 0.011	0.698 ± 0.023
GroupNorm	G4: Vanilla ER	41.2 ± 13.5	34.9 ± 23.7	0.730 ± 0.011	0.309 ± 0.111
	G6: Curriculum ($\lambda_{\min}$=0.20)	$\mathbf{5.2 \pm 2.1}$	$\mathbf{1.7 \pm 0.9}$	0.731 ± 0.020	0.670 ± 0.055
	G8: Asym. PER (δ=0.1)	5.6 ± 1.1	4.8 ± 3.8	0.741 ± 0.009	0.673 ± 0.023

Table 7.6: CIFAR-10 headline block (ResNet-18, momentum 0.9, buffer 5,000, five seeds). Both gates cut gap depth and area far below vanilla ER at matched final accuracy and much higher worst-case retention (min-ACC), under both normalisations. Neither was given a CIFAR-specific sweep; both carry their rot-MNIST configurations.

stays open for hundreds of steps, so the area carries as much of the comparison as the depth (Figure 7.5): vanilla ER under group norm crashes T_0 almost to chance and crawls back over the whole recovery window, while both gates hold T_0 near its plateau through the switch. Depths are directly comparable within this block; areas only within it, since the CIFAR window is several times the rot-MNIST one.

The mechanism partially generalises. The momentum cross repeats the composability test of Section 6.2.3 on the deep backbone, pairing momentum-0 runs at group norm against the momentum-0.9 rows (Table 7.7). Two of the three predicted signatures reproduce: momentum leaves vanilla ER's depth essentially unchanged (40.5 → 41.2 pp, the deterministic-first-step null), and it composes with the curriculum, taking depth from 28.6 to 5.2 pp (source attenuation). The third, momentum re-inflating the directional gate, does not appear: momentum takes asymmetric PER from 22.6 to 5.6 pp instead. We read this as a convergence confound rather than a failure of the account. At momentum 0 the ResNet under-converges the past task in ten epochs (vanilla min-ACC 0.11, ACC 0.65), so the momentum-0 depths sit on a degraded baseline and the momentum contrast for the directional gate is dominated by the general convergence gain rather than by the per-step-attenuation effect the full-buffer MLP isolates cleanly. The per-step re-inflation signature is established on rot-MNIST (Section 7.3); on this backbone we can confirm the mechanism only for the two gates whose momentum contrast is not swamped by convergence.

Method (GroupNorm)	depth, μ=0	depth, μ=0.9
Vanilla ER	40.5 ± 4.0	41.2 ± 13.5
Curriculum ($\lambda_{\min}$=0.20)	28.6 ± 8.8	5.2 ± 2.1
Asym. PER (δ=0.1)	22.6 ± 9.7	5.6 ± 1.1

Table 7.7: CIFAR-10 momentum cross (group norm, buffer 5,000, five seeds), gap depth in pp. The vanilla null and the curriculum composition reproduce; the directional-gate re-inflation does not, confounded by momentum-0 under-convergence of the ResNet (see text).

The curriculum generalises to unseen corruptions, with one reliability boundary. A further block pairs the shipping curriculum against a vanilla-ER control on shifts it was not tuned on (three seeds, group norm; Table 7.8). On the two novel two-task corruptions the curriculum generalises cleanly, cutting depth from 46.3 to 7.3 pp on shot noise and from 8.4 to 3.9 pp on contrast. The three-task sequence is the exception: there the curriculum underperforms its control (32.8 vs. 4.4 pp, seed std ±14.5), an anomaly localised entirely to the Gaussian → shot transition between two near-identical corruptions and absent from the batch-norm variant of the same sequence (Appendix C.5). At that boundary the incoming task is already partly solved, so $\|g_{\text{new}}\|$ is small and the adaptive ratio $r = \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ loses conditioning; both norms are small rather than the small-replay/large-new regime the schedule was built for, and λ tracks erratically, consistent with the large seed spread. We flag it as a reliability boundary of the adaptive ratio under near-identical consecutive shifts, not as evidence against the two-task result: final accuracy stays comparable to vanilla throughout and the failure is seed-unstable rather than systematic.

Shift (GroupNorm)	Curriculum depth	Vanilla depth
2-task shot noise	$\mathbf{7.3 \pm 5.9}$	46.3 ± 12.0
2-task contrast	$\mathbf{3.9 \pm 3.1}$	8.4 ± 4.4
3-task g.n.+shot	32.8 ± 14.5	$\mathbf{4.4 \pm 0.4}$

Table 7.8: CIFAR-10 generalisation block, gap depth (pp), curriculum versus paired vanilla-ER control (three seeds). The curriculum wins on both novel two-task corruptions; the three-task group-norm sequence is an outlier.

Compute cost separates the two gates where stability and plasticity do not. The frontier ranks the gates on retention and accuracy; compute is a third axis, and on it the two gates are not close. The scalar gate is a single changing scalar — one gradient-norm ratio per step, no stored curvature — so its wall-clock is indistinguishable from vanilla ER: on the ResNet-18 the gate-active task (T_1) takes 2,833 s under the curriculum against 2,836 s under vanilla at batch norm, and 4,749 vs. 4,624 s at group norm (Table 7.9). The directional gate runs 25 conjugate-gradient Fisher-vector products per step, each on the order of a backward pass, and

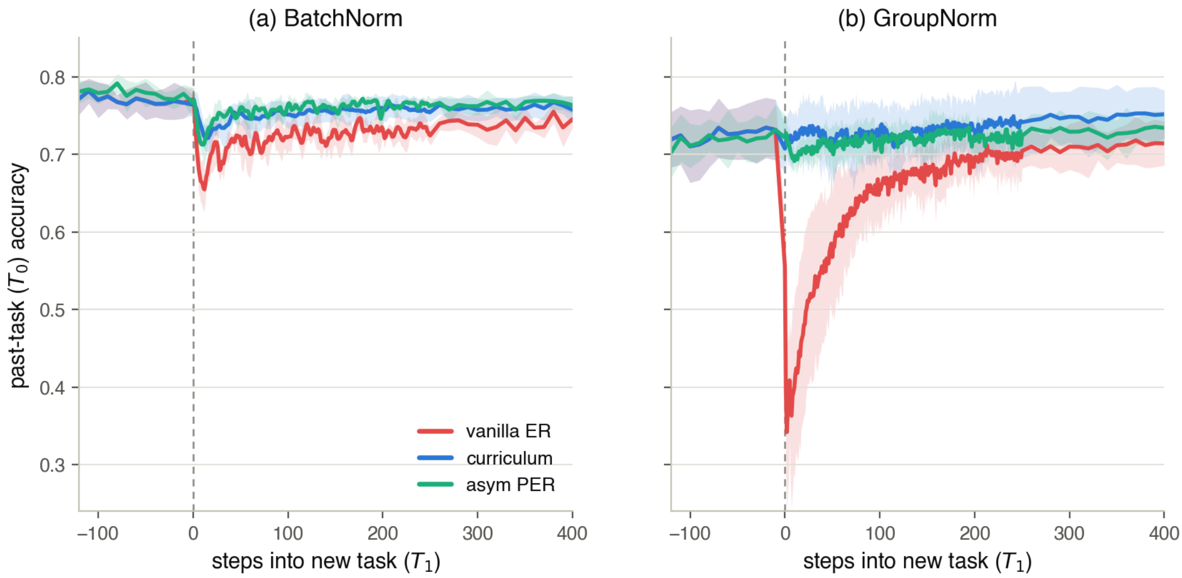

Figure 7.5: Per-step past-task (T_0) accuracy across the CIFAR-10 transition (ResNet-18, momentum 0.9, buffer 5,000), one panel per normalisation, each overlaying vanilla ER (red), the shipping curriculum (blue) and asymmetric PER (aqua); seed mean with ±std bands, x=0 the task switch. (a) BatchNorm: vanilla ER dips and re-adapts slowly while both gates hold T_0 near its plateau. (b) GroupNorm: vanilla ER crashes T_0 almost to chance and crawls back over the whole recovery window, whereas both gates hold it flat through the switch — cutting the deep transient and lifting worst-case retention from ≈0.31 to ≈0.67. On this backbone the transient stays open for hundreds of steps, so the area carries as much of the comparison as the depth (Table 7.6, Appendix C.4).

the cost shows on the deep backbone: the same task takes 11,712 s (batch norm) and 10,057 s (group norm), a 4.1× and 2.1× multiple of the curriculum. Read as the slowdown of the gate-active task relative to the un-gated first task on the same hardware — a machine-independent ratio — vanilla and the curriculum both sit at ≈5× while asymmetric PER sits at 24–29×. On the MLP the same overhead is only ≈8% (curriculum 231 s, PER 250 s per step-block), because there the cheap backward pass and the per-step evaluation dominate the wall clock and mask the solve; the gap widens with backbone depth, exactly as the per-step-cost argument predicts. The scalar gate is therefore the cheaper recommendation for systems queried during training, and the directional gate the choice where its per-step Fisher cost is affordable.

Gate (task T_1, gate active)	rot-MNIST MLP	ResNet-18 BN	ResNet-18 GN
Vanilla ER	—	2,836 s	4,624 s
Curriculum (scalar)	231 s	2,833 s	4,749 s
Asym. PER (δ=0.1)	250 s	11,712 s	10,057 s

Table 7.9: Wall-clock of the gate-active task (T_1), mean over five seeds. The scalar gate's cost is indistinguishable from vanilla ER; the directional gate's per-step conjugate-gradient solve multiplies it up to 4× on the ResNet-18, and the multiple grows with backbone depth. The MLP vanilla cell is omitted because its driver-block run additionally carries the per-step gradient-fidelity diagnostic, which would inflate it; the curriculum and PER MLP runs carry no such diagnostic and are directly comparable.

7.6 Summary of Findings

The ladder holds end to end. (i) The gap separates into a deterministic arc and overshoot of the followed path: removing the drivers leaves the arc (D4: 1.9 pp depth, area 3.2), fixing the path leaves the overshoot (C7: 13.9 pp), doing both closes the gap (C8: 0.2 pp), while the permanent stability–plasticity cost stays on the ledger (C8 ACC 0.818). (ii) The scalar feedback gate times the push down the valley-floor path and, with momentum, reaches 3.7 pp of depth at vanilla accuracy; momentum closes its gap by averaging variance and pays back the plasticity the schedule borrows, so the curriculum needs momentum for accuracy, not for the gap. (iii) The directional feedforward gate shrinks depth and arc monotonically as $\delta \to 0$ at no plasticity cost and dominates the scalar gate at momentum 0, but has curvature blind spots (the δ=0.03 cliffs, solver-controlled) and re-inflates under momentum. (iv) The two gates trace a stability–plasticity frontier rather than an ordering, and both legs carry to a ResNet-18 on CIFAR-10 under both normalisations at matched accuracy and much improved worst-case retention; the vanilla and curriculum momentum signatures carry too, with the directional re-inflation confounded by under-convergence on the deep backbone, and a single three-task group-norm sequence flagged as a reliability boundary of the adaptive ratio. Compute is the third axis the frontier hides: the scalar gate costs no more than vanilla ER, while the directional gate's per-step Fisher-vector solve runs up to 4× slower on the ResNet-18 (Table 7.9). The discussion reads these findings against the account of Sections 3 and 4.

8 Discussion

The results defend the two-level account and, with it, the gate view of replay. This section reads them back against the theory: what the two levels explain that a single additive decomposition could not (Section 8.1), how the two gates divide the stability–plasticity work along the feedback/feedforward and scalar/directional axes (Section 8.2), the limits of the evidence (Section 8.3), and where the taxonomy points next (Section 8.4).

8.1 Why the gap needs two levels

The central empirical claim is that the stability gap is not one quantity but two failures of the same optimisation, kept apart because they answer to different interventions. The 2×2 of Section 7.1 is the direct test. Removing the overshoot drivers by construction leaves a deterministic arc (D4: 1.9 pp of depth but a clear bend, area 3.2, with the replay gradient exact and the buffer noiseless); fixing the path with the curriculum leaves the overshoot (C7 at momentum 0: 13.9 pp, essentially all of it); only doing both closes the gap (C8: 0.2 pp). Neither intervention is a weaker version of the other. The arc is a property of which path the descent dynamics define, and no amount of optimiser hygiene, exact gradients, balanced magnitudes, zero noise, touches it; the overshoot is how far the discrete, noisy, accelerated iterate leaves whatever path it is on, and no change of path removes it. This is why the levels must be named separately: an intervention aimed at one is silent on the other.

This supersedes the additive decomposition an earlier version of this work reported, and the reason it does is instructive rather than cosmetic. A partition into a magnitude share, a variance share, and a trajectory share fails three ways that the two-level account does not. It is order-dependent: the estimator "share" measured by introducing the exact gradient before balancing differs from the share measured after it, because a noisy replay gradient also inflates the first-step magnitude, so the factors are not separable summands. It maps factors to metrics that the data then crosses: the curriculum's momentum-0 tail looks like the "trajectory" term until C8 removes it with the exact buffer and no momentum at all (Table 7.4), revealing it as variance; the same metric is produced by different causes in different conditions. And it is a category error: a boundary imbalance is a state, estimator variance is a property of the replay sample, and the arc is a property of the vector field, and summing three things of different kinds mistakes drivers of a process and the shape of a path for parts of one number. The two-level split is by kind, which path is followed versus how tightly it is followed, and that is the distinction the interventions actually respect.

The split also settles what the gap is and is not the price of. Reaching the joint optimum requires the replay loss to rise to its joint-basin level, and that rise is permanent; it is the stability–plasticity cost, and C8 shows it plainly, closing the transient to 0.2 pp while its final accuracy sits 5.5 pp below vanilla ER (Section 7.1). Everything the accuracy curve does below its settled level and back up again is failure to follow the monotone reference path of Section 3.1, not the cost of learning the new task. Keeping the transient (the gap) apart from the permanent accommodation (forgetting) is what lets the end-referenced area attribute the gap to the trajectory rather than to capacity loss, and it is why the area is referenced to the recovered plateau by construction rather than to the pre-switch baseline.

Finally, the two levels explain why the metric that matters changes with the backbone. On the MLP the arc is cheap: the network reconverges in a few steps, so the open bend integrates to little and the perceived severity is the overshoot spike. On the ResNet the same transient stays open for hundreds of steps (Figure 7.5), so the arc, the part that persists, is where most of the accuracy the gap costs accumulates, and it is where both gates' margin over vanilla is largest (Table 7.6). Depth and area are not two views of one number; they are the signatures of the two levels, and which one dominates the cost is set by how slowly the network recovers. A depth-only reading would have understated the result on exactly the architectures where continual-evaluation cost is highest.

8.2 Two gates, two ways to divide the work

Written in the unified update of Section 3.4, every replay method gates the new-task push while restoring at full strength, and the curriculum and asymmetric PER are the two pure corners of the taxonomy: a scalar gate driven by feedback, and a directional gate driven by feedforward. The results show the two corners working and failing exactly where the taxonomy says they should, and the single fact that organises the whole comparison is *where* the gate attenuates the push.

The scalar feedback gate attenuates at source. It multiplies the entire new-task loss by a schedule that starts near zero, so early in the transition there is almost no push to discretise, accumulate, or let noise act on, and by the time the weight has grown the trajectory has already settled toward the joint basin. This has three consequences the data confirm. It composes with momentum: because the push is small and clean at source, momentum finds a stochastic residual to average and closes the gap it left untouched under vanilla ER (Section 7.2), while the accumulator has no sustained full-strength direction to rebuild. It is blind-spot-proof: the adaptive schedule reads the live replay-gradient norm, a signal that fires whatever direction the damage arrives from, so it cannot be steered past a threat it did not model. And it pays a plasticity debt: holding the loss down slows the new task, which is why C8 at momentum 0 is gap-free but 5.5 pp down on accuracy, and why the debt is repaid only by acceleration, momentum lifting C8 to the highest accuracy in the benchmark (Table 7.4). The scalar gate's price is plasticity, and momentum is how it is paid.

The directional feedforward gate attenuates per step. It filters the push through the replay Fisher every step, passing motion where the past task is flat and braking it where the past task is curved, and it never slows the new task at source, so it protects the past task at no plasticity cost, ACC and short-horizon plasticity rising as the filter strengthens (Table 7.5). The same per-step locality is its weakness on both of the taxonomy's predicted failure axes. Under momentum the protection unravels: the driving magnitude of the new-task gradient is untouched, so velocity accumulated across steps reconstructs displacement in the very directions the filter shrinks one step at a time, and depth re-inflates from ≈ 2 to ≈ 8 pp (Section 7.3). This is the same lever as the curriculum, momentum, producing the opposite outcome, and the difference is entirely one of where the attenuation was applied: source versus step. That single distinction accounts for the whole momentum column of the study, the vanilla depth null (no accumulated velocity in the deterministic first step), the curriculum's composition (source attenuation), and the directional gate's re-inflation (step attenuation).

The directional gate's second failure is the sharper one, because it is the case for feedback. At the strongest filter its curvature model has blind spots: the sampled Fisher rates a direction flat that the true past-task loss punishes further out, new-task pressure escapes through it, and the past-task curve slides off a late cliff (the δ=0.03 full-buffer cliffs, reproduced under a stronger solver, Section 7.3). A feedforward gate can only be as good as its model of the past task, and that model is estimated on finite data under a saturating loss. A feedback gate has no such blind spot, because it does not model the damage in advance, it reads it. The cliffs and the plasticity debt are therefore the two gates' defining costs, and they are complementary: the feedforward gate is cheap in plasticity but myopic, the feedback gate is blind-spot-proof but slow. This is why the two do not order but trace a frontier (Section 7.4): at momentum 0 the directional gate dominates because it pays no plasticity and momentum has not yet unwound its filter, while under momentum the scalar gate reaches the lowest depth and the directional gate the highest accuracy, and the right choice depends on whether worst-case retention during the transition or peak final accuracy is the operative cost. That both legs carry to the ResNet at matched accuracy and much improved worst-case retention (Table 7.6), and that the two mechanism signatures which are not confounded by convergence reproduce there (Table 7.7), is the evidence that it is the gate principle, not either particular method, that generalises.

8.3 Limitations

Several limits bound these claims. The first is statistical: five seeds floor the two-sided Wilcoxon test at $p = 0.0625$, so the argument rests on effect sizes and seed-paired direction, not a 0.05 threshold the design cannot reach. The effects the story leans on, the order-of-magnitude depth reductions on CIFAR, the C7→C8 variance collapse, the momentum re-inflation of the directional gate, are large relative to the seed spread and unanimous across seeds; the smaller ones, the adjacent-δ steps and the $\lambda_{\min}$ trade-offs, are read as directional consistency only, and one, the adaptive-versus-best-linear depth, does not reach even that ($p = 0.44$).

The second is the accuracy drift of the scalar gate at momentum 0: lengthening the linear ramp lowers depth but costs several points of final accuracy (Table 7.3), so the method's accuracy-neutrality is a property of the full curriculum-plus-momentum configuration, not of the ramp alone. The third is the directional gate's cliffs, which we establish at the blind-spot level the solver control supports; the finer softmax-saturation account of why they land on the full-buffer leg is flagged as interpretation, not evidence. The fourth is the CIFAR momentum cross: the directional-gate re-inflation signature does not reproduce on the ResNet, and while we attribute this to momentum-0 under-convergence of the backbone rather than to a failure of the mechanism (which the MLP isolates cleanly), a clean deep-backbone test of that one signature would require a converged momentum-0 baseline we did not run. The fifth is the three-task group-norm sequence, where the adaptive ratio loses conditioning at a boundary between two near-identical corruptions and the schedule underperforms its control with high seed variance; we flag it as a reliability boundary of the gradient-ratio feedback signal, absent from the batch-norm variant, rather than a property of the gate principle.

Two scope limits are worth stating plainly. The full-buffer legs of both the driver block and the δ sweep store the entire past task and so break the limited-memory premise of continual learning; they are diagnostic probes that isolate the arc and split momentum's roles, and the practical configurations are the 1k-buffer curriculum ($C7_M$) and the standard-buffer directional gate. And the directional gate's per-step Fisher-vector solve adds a fixed multiple of the backward cost per step (25 conjugate-gradient iterations here), a real overhead on large backbones — up to $4\times$ the gate-active wall-clock of the curriculum on the ResNet-18 (Table 7.9) — that the scalar gate, a single changing scalar with no stored curvature, does not carry: its cost is indistinguishable from vanilla ER. The headline analysis also rests on a single-transition, two-task design chosen to isolate one boundary; a systematic study across many boundaries, longer sequences, and class-incremental rather than domain-incremental shifts is left open.

8.4 Outlook

The taxonomy that organises the two methods also names what is missing from it. The two gates studied here occupy two of its four corners, scalar×feedback and directional×feedforward, and the empty corner, a directional gate driven by a live damage signal rather than by a precomputed curvature model, is exactly the design the cliffs argue for: it would filter the push by direction, keeping the plasticity that makes the feedforward gate cheap, while reading the damage as it arrives, closing the blind spot that makes it myopic. Building and testing such a feedback-directional hybrid is the most direct consequence of these results.

Two narrower directions follow from the failure modes. The directional gate's momentum reinflation is a per-step-attenuation artefact; a version that attenuates the accumulated velocity rather than the instantaneous step, or that damps the push at source in the directions the Fisher flags, would be predicted to compose with momentum the way the scalar gate does, a testable claim. And momentum's gap-closing role was shown to be interchangeable with source variance reduction (C8), which suggests that iterate averaging, a variance reducer that carries none of momentum's accumulator-driven overshoot, could close the stochastic residual without the small transient price acceleration re-introduces. On the practical side, the scalar gate is the cheaper recommendation for systems queried during training, closing the gap at almost no final-accuracy penalty and negligible overhead; the directional gate is the choice where peak accuracy is the operative metric and the per-step Fisher cost is affordable. The frontier, not a single method, is what this work leaves for a practitioner to choose along.

Statement on the Use of AI Tools

In accordance with the University of Groningen's guidelines on the responsible use of generative AI, I disclose that AI assistants were used during the preparation of this thesis. The tools were Google Gemini Pro (version 3 Pro, then version 3.1 Pro from February 2026) and Anthropic Claude Opus (the 4.x line, versions 4.5 through 4.8) — the releases current across the roughly six-month period, January to July 2026, in which the work was carried out. They functioned as assistive tools under my continuous direction, not as autonomous contributors. Specifically, I used them to:

- brainstorm and stress-test ideas, and as a sounding board to reason through arguments and explanations;
- draft and revise prose, which I subsequently edited for accuracy, precision, and voice;
- assist in writing and refactoring the analysis and experiment code, all of which I reviewed, ran, and tested;
- produce first-pass summaries of the literature, which I verified against the primary sources.

The tools were never given open-ended control: every request specified a concrete, bounded task, and all output was reviewed and, where retained, corrected, understood, and integrated by me. All experimental results, figures, and numerical claims were checked against the underlying run data, and all cited work was read in the original. The research questions, experimental design, interpretation of the results, and the overall argument of the thesis are my own, and I take full responsibility for its content, including any errors.

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A Implementation Details

This appendix records the implementation specifics that the methods of Section 6 depend on but that would interrupt the argument in the main text: the experience-replay variants, and the NCL realisation together with the hyperparameters fixed by the rot-MNIST tuning sweep.

A.1 Datasets

rot-MNIST rotates each 28×28 image by the per-task angle and feeds the flattened greyscale vector to the MLP. The CIFAR-10 domain shifts are generated with the `imagecorruptions` package, which implements the standard common-corruptions benchmark of Hendrycks and Dietterich [10]; the Gaussian noise, shot noise, and contrast shifts are applied at severity 3 on the benchmark's 1–5 scale, per sample at load time, followed by the standard CIFAR-10 channel normalisation. The clean task ("none") applies the normalisation only.

A.2 Models and Training

Tables A.1 and A.2 give the two backbones, and Table A.3 the training protocol shared across both benchmarks.

Component	Value
Input	784 (flattened 28×28 greyscale)
Hidden layers	2×400 (fully connected, ReLU)
Output	10-class logits
Regularisation	none
Total parameters	478,410

Table A.1: MLP backbone used for the rot-MNIST sweeps.

Component	Value
Input	$3 \times 32 \times 32$ RGB
Stem	3×3 conv, stride 1, no max-pool (CIFAR adaptation)
Stages	4 residual stages, BasicBlocks $[2, 2, 2, 2]$, widths $[64, 128, 256, 512]$
Normalisation	BatchNorm (default); GroupNorm ($\min(32, C)$ groups) variant crossed in Section 6.2.5
Head	global average pool + linear, 10-class logits
Total parameters	$\approx$11.2M

Table A.2: CIFAR-adapted ResNet-18 backbone used for the CIFAR-10 block. The 3×3 stride-1 stem and removed max-pool keep the feature map at 32×32; all other details follow the ResNet-18.

Hyperparameter	rot-MNIST (MLP)	CIFAR-10 (ResNet-18)
Epochs per task	1 (online)	10
Batch size	256	256
Optimiser	SGD	SGD
Learning rate	0.1	0.1
Momentum	0.0 or 0.9	0.9 (0.0 in cross)
Weight decay	0.0	0.0
Replay buffer	1,000 or 60,000	5,000
Seeds	5	5 or 3

Table A.3: Training protocol for both benchmarks.

A.3 Experience replay

The replay buffer uses reservoir sampling [25], so the stored set is a uniform sample of the past stream at any budget. Three modes appear in the experiments:

- *Standard ER* (the D1 and all C-series conditions): each step takes one combined backward pass on $L_{\text{current}} + L_{\text{replay}}$, with half the mini-batch drawn from the current task and half sampled with replacement from the buffer.
- *Balanced ER* (D3): two separate backward passes whose gradients are individually unit-normalised before being combined 50/50, removing the magnitude asymmetry between g_{new} and g_{replay}.
- *Full-data replay* (D2, D4): the replay gradient is computed on *every* stored sample exactly once per step (no sub-sampling). Paired with a buffer equal to the full past-task training set (60,000 for rot-MNIST T_0), this yields the deterministic empirical past-task gradient at the current parameters, removing estimator variance.

The λ-curriculum re-weights only the current-task term, minimising $L_\lambda = L_{\text{replay}} + \lambda(t)\, L_{\text{current}}$; the adaptive schedule reads $\|g_{\text{replay}}\|/\|g_{\text{new}}\|$ through an exponential moving average ($\alpha = 0.05$) and applies the optional floor $\lambda_{\min}$.

A.4 Asymmetric PER

The filtered push $\delta\,(F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}}$ of Equation (1.2) is computed each step by warm-started conjugate gradients over Fisher-vector products (25 iterations), so the metric is never formed as a matrix; warm-starting from the previous step's solution keeps the per-step solve cheap across the run. The solver control of Section 6.2.4 reruns the strongest filter ($\delta = 0.03$, full buffer) at 60 iterations to separate properties of the gate from artefacts of an under-converged solve. On the full-buffer leg the replay gradient is exact over all 60k past samples while the Fisher stays estimated on a sampled batch, since a full-buffer Fisher-vector product per CG iteration would dominate the runtime.

A.5 Natural Continual Learning

NCL follows Kao et al. [14] (their Eq. 8 and Algorithm 1). The update preconditions the new-task gradient by the prior covariance Λ_{k-1}^{-1} and adds the Bayesian restoring term,

$$\theta \leftarrow \theta - \eta \left[\Lambda_{k-1}^{-1} \nabla L_k(\theta) + (\theta - \mu_{k-1})\right],$$

with Λ_{k-1} summarised by a K-FAC factorisation $F \approx A \otimes G$ per linear layer and an identity prior $p_w = \alpha\, I$. The trust radius of the paper's Eq. 7 is absorbed into the learning rate; we log a `tr_scale` diagnostic but do not clip the step.

Tuned hyperparameters. The values used in every NCL condition are those that won the rot-MNIST tuning sweep below (Table A.4). The paper-literal prior $\alpha = 1.0$ over-regularises on rot-MNIST and the damping ε is a numerical safeguard only: the identity prior, not the damping, is what bounds Λ^{-1}. These values were tuned on the two-task rot-MNIST/MLP setting and reused unchanged on the CIFAR-10 ResNet-18, as we did not run a separate NCL sweep on the deeper backbone. NCL's CIFAR results should therefore be read as those of a method tuned on the smaller benchmark rather than re-optimised for CIFAR, a caveat we revisit in Section 8.3.

Hyperparameter	Symbol	Value
Prior precision	α (`prior_init`)	0.1
Damping	ε (`damping`)	1e−3
Fisher samples (K-FAC)	`fisher_samples`	1,000
Trust radius (diagnostic)	r (`trust_radius`)	1.0
Learning rate	η	0.1
Momentum	μ	0.9

Table A.4: NCL hyperparameters fixed for all conditions, from the rot-MNIST tuning sweep of Table A.5.

Tuning sweep. The prior precision α was tuned on the two-task rot-MNIST/MLP setting (Table A.5), holding damping at 1e−3 and `fisher_samples` at 1,000. Smaller α improves over the paper-literal $\alpha = 1.0$, but the gain saturates: $\alpha = 0.1$ gives the best stability–accuracy trade-off, while $\alpha \leq 0.03$ becomes unstable at $\eta = 0.1$ (the natural gradient amplifies the raw gradient by up to $1/\alpha$ in any direction), and halving the learning rate does not recover it.

Config	α	η	Momentum
alpha_0.1 (chosen)	0.1	0.1	0.9
alpha_0.03	0.03	0.1	0.9
alpha_0.01	0.01	0.1	0.9
alpha_0.003	0.003	0.1	0.9
alpha_0.03_lowlr	0.03	0.05	0.9

Table A.5: NCL prior-precision tuning sweep on two-task rot-MNIST. Damping (1e−3) and `fisher_samples` (1,000) held fixed.

B Metric Definitions

This appendix gives the full definitions of every metric recorded by the evaluation pipeline of Section 6.1.3. The main text explains the measures the argument turns on; the tables below are the look-up reference for the complete set.

Stability-gap metrics. Computed from the dense per-step evaluation. Write a_j^{pre} for the pre-task accuracy on task j and $a_j(t)$ for its accuracy at step t of the new task.

Metric	Definition	Role
`gap_depth`	$\max_j \big(a_j^{\mathrm{pre}} - \min_t a_j(t)\big)$ over the full record window	Primary metric, worst single point of instability
`gap_area_end`	$\sum_j \int_0^T \max(0,\, b_j - a_j(t))\, dt$, trapezoidal over the full new-task record (to task end T), with $b_j = \min\big(a_j^{\mathrm{pre}},\, \bar{a}_j^{\mathrm{end}}\big)$ and $\bar{a}_j^{\mathrm{end}}$ the trailing-window mean (last $\max(5, \lceil 0.1n \rceil)$ samples); accuracy at or above b_j contributes 0	Transient cost reported in the results: dip below the recovered level until it closes, excluding permanent forgetting and continued learning (units: accuracy · steps)
`recovery_steps`	Given a record has dipped below the threshold, the first step (counted from the transition) at which *every* past task is simultaneously at $\geq 90\%$ of its pre-task baseline	None if no sub-threshold dip occurs, or if recovery is not reached within the record
`gap_area`	As `gap_area_end` but referenced to the pre-task baseline throughout: $\sum_j \int_0^T \max\big(0,\, a_j^{\mathrm{pre}} - a_j(t)\big)\, dt$	Pre-referenced area; bundles the transient dip with permanent below-baseline drift. Superseded by `gap_area_end`, logged for back-comparability
`max_drop`	`gap_depth` restricted to the first 200 steps	Legacy fixed-window depth, kept for back-comparability

Table B.1: Stability-gap metrics, recorded from dense per-step evaluation. `gap_depth` and `gap_area_end` are the two axes used in the main analysis, how deep the gap cuts and how long it stays open, together with `recovery_steps`; the rows below the rule are secondary or legacy measures logged by the pipeline but not reported in the main text.

Continual-learning metrics. Following Lange et al. [16], computed at task boundaries and over the full record. Let $A(E_j, f_n)$ be the accuracy on the test set of task j evaluated at the model f_n after training stage n, and let N be the number of tasks. The full task-boundary accuracy matrix $R[i,j] = A(E_j, f_{T_i})$ is logged for every run, so any downstream metric can be reconstructed.

Metric	Definition	Role
ACC	$\frac{1}{N}\sum_j A(E_j, f_{T_{N-1}})$	Average task accuracy at the final model
FORG	$\frac{1}{N-1}\sum_{i<N-1} \big[A(E_i, f_{T_i}) - A(E_i, f_{T_{N-1}})\big]$	Average drop from right-after-learning to final
min-ACC	$\frac{1}{N-1}\sum_{i<N-1} \min\{A(E_i, f_n) : n > T_i\}$	Worst-case retention since learning
WF_w	Mean over tasks of the largest accuracy drop in any window of w consecutive evaluations ($w \in \{10, 100\}$)	Worst-case stability over short / medium horizons
WP_w	Symmetric counterpart of WF_w: the largest accuracy gain ($w \in \{10, 100\}$)	Plasticity / learning velocity
WC-ACC	$\frac{1}{N}A(E_{N-1}, f_{T_{N-1}}) + \big(1 - \frac{1}{N}\big)$ min-ACC	Worst-case trade-off — a lower bound on ACC

Table B.2: Continual-learning metrics of Lange et al. [16], recorded at task boundaries and over the full record.

Gradient-fidelity diagnostics. Recorded at the dense cadence for the decomposition conditions only. The tracker compares the replay-buffer gradient g_{replay} against the true past-task gradient g_{true}, computed at every step by a separate forward and backward pass over the entire past-task training set (no sub-sampling, so g_{true} is the deterministic empirical past-task gradient at the current parameters).

Diagnostic	Logged name	Role
$\cos(g_{\text{replay}}, g_{\text{true}})$	`true_grad_cosine`	Buffer-side directional fidelity, the estimator-variance driver of Section 6.2.1
$\lVert g_{\text{replay}} \rVert / \lVert g_{\text{true}} \rVert$	`true_grad_mag_ratio`	Buffer-side magnitude fidelity
$\lVert g_{\text{new}} \rVert / \lVert g_{\text{replay}} \rVert$	`grad_ratio`	Update-side magnitude asymmetry, large at step 1 when $\lVert g_{\text{replay}} \rVert \approx 0$

Table B.3: Gradient-fidelity diagnostics. Each is logged as a per-step series together with its mean (and minimum, for the cosine). The first two carry the `true_grad` prefix because they are measured against g_{true}; `grad_ratio` compares g_{replay} to the current-task gradient g_{new} and its inverse drives the adaptive curriculum of Section 6.2.2.

C Complete Results

This appendix gives the full metric set for every condition, as recorded by the aggregation pipeline of Section 6.1.3. Each row is the mean over five seeds (three for the CIFAR generalisation block) with the seed standard deviation. Accuracy-type metrics (ACC, FORG, min-ACC, WF_{100}, WP_{100}, WC-ACC) are fractions; `gap_depth` is in percentage points (pp); `gap_area_end` is the transient cost (accuracy-fraction $\times$ steps, Table B.1), the dip below the level each task recovers to, so permanent forgetting does not contribute; "Recov." is `recovery_steps`, with "—" marking conditions where every past task stayed above the 90% recovery threshold for the whole window (no recovery event to time).

C.1 rot-MNIST: Decomposition (D-series)

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end	Recov.
Momentum 0.0									
D1 vanilla	0.873 ± 0.005	0.019 ± 0.020	0.706 ± 0.056	0.147 ± 0.029	0.736 ± 0.020	0.794 ± 0.028	19.5 ± 4.1	6.3 ± 0.5	3.6 ± 0.9
D2 fulldata	0.890 ± 0.009	-0.016 ± 0.009	0.719 ± 0.039	0.126 ± 0.020	0.736 ± 0.018	0.801 ± 0.022	18.3 ± 2.5	8.1 ± 0.7	3.0 ± 1.2
D3 balanced	0.849 ± 0.005	0.031 ± 0.018	0.824 ± 0.006	0.045 ± 0.002	0.721 ± 0.004	0.836 ± 0.004	5.7 ± 1.9	1.4 ± 0.4	—
D4 fulldata balanced	0.869 ± 0.002	-0.014 ± 0.013	0.862 ± 0.002	0.030 ± 0.007	0.710 ± 0.003	0.852 ± 0.002	1.9 ± 1.3	3.2 ± 0.5	—
D7 NCL reference	0.770 ± 0.012	0.057 ± 0.013	0.809 ± 0.022	0.109 ± 0.023	0.639 ± 0.012	0.762 ± 0.011	7.2 ± 1.0	0.3 ± 0.1	125.0
Momentum 0.9									
D1 vanilla	0.928 ± 0.003	0.058 ± 0.007	0.797 ± 0.028	0.092 ± 0.016	0.808 ± 0.012	0.878 ± 0.014	15.9 ± 2.6	1.0 ± 0.2	12.6 ± 1.7
D2 fulldata	0.964 ± 0.002	-0.010 ± 0.003	0.807 ± 0.009	0.082 ± 0.004	0.808 ± 0.012	0.885 ± 0.004	14.9 ± 0.9	3.1 ± 0.2	10.6 ± 1.1
D3 balanced	0.932 ± 0.005	0.038 ± 0.003	0.897 ± 0.006	0.040 ± 0.003	0.822 ± 0.011	0.922 ± 0.003	5.9 ± 0.6	0.4 ± 0.2	—
D4 fulldata balanced	0.967 ± 0.001	-0.022 ± 0.003	0.942 ± 0.005	0.016 ± 0.004	0.834 ± 0.010	0.950 ± 0.002	1.3 ± 0.4	0.2 ± 0.1	—
D7 NCL reference	0.810 ± 0.019	0.021 ± 0.004	0.871 ± 0.012	0.218 ± 0.014	0.686 ± 0.012	0.778 ± 0.022	8.5 ± 1.1	0.6 ± 0.1	5.0

Table C.1: rot-MNIST decomposition block, full metrics, at both momentum settings. Buffer-fidelity diagnostics for these runs are in Table C.3.

C.2 rot-MNIST: λ-Curriculum (C-series)

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end	Recov.
Momentum 0.0									
C1 linear N50	0.851 ± 0.023	0.042 ± 0.021	0.707 ± 0.019	0.148 ± 0.011	0.748 ± 0.006	0.784 ± 0.021	17.5 ± 1.7	4.9 ± 0.5	35.8 ± 6.3
C2 linear N100	0.844 ± 0.025	0.046 ± 0.023	0.719 ± 0.033	0.140 ± 0.025	0.730 ± 0.006	0.786 ± 0.019	16.2 ± 4.3	3.6 ± 0.3	64.0 ± 8.6
C3 linear N200	0.828 ± 0.025	0.054 ± 0.024	0.739 ± 0.048	0.128 ± 0.043	0.688 ± 0.004	0.784 ± 0.034	14.3 ± 4.6	1.8 ± 0.6	132.8 ± 43.4
C4 adaptive	0.833 ± 0.025	0.050 ± 0.019	0.754 ± 0.025	0.111 ± 0.016	0.703 ± 0.007	0.794 ± 0.020	12.8 ± 2.9	2.2 ± 0.5	88.8 ± 6.1
C5 adaptive lmin0.05	0.835 ± 0.025	0.048 ± 0.019	0.749 ± 0.027	0.113 ± 0.016	0.702 ± 0.006	0.793 ± 0.020	13.2 ± 3.2	2.2 ± 0.5	88.8 ± 6.1
C6 adaptive lmin0.10	0.836 ± 0.026	0.048 ± 0.021	0.743 ± 0.026	0.120 ± 0.019	0.700 ± 0.007	0.791 ± 0.019	13.9 ± 3.3	2.4 ± 0.5	88.4 ± 11.0
C7 adaptive lmin0.20	0.837 ± 0.031	0.049 ± 0.024	0.743 ± 0.030	0.119 ± 0.023	0.710 ± 0.010	0.792 ± 0.024	13.9 ± 3.4	2.7 ± 0.6	64.8 ± 8.2
Momentum 0.9									
C1 linear N50	0.932 ± 0.003	0.053 ± 0.005	0.892 ± 0.004	0.041 ± 0.004	0.830 ± 0.012	0.927 ± 0.003	6.4 ± 0.5	0.2 ± 0.1	—
C2 linear N100	0.932 ± 0.003	0.051 ± 0.008	0.892 ± 0.005	0.036 ± 0.003	0.826 ± 0.012	0.926 ± 0.002	6.4 ± 0.6	0.2 ± 0.1	—
C3 linear N200	0.925 ± 0.007	0.051 ± 0.006	0.893 ± 0.004	0.031 ± 0.004	0.819 ± 0.010	0.920 ± 0.006	6.2 ± 0.6	0.1 ± 0.0	—
C4 adaptive	0.917 ± 0.001	0.019 ± 0.004	0.931 ± 0.005	0.016 ± 0.003	0.798 ± 0.010	0.915 ± 0.001	2.5 ± 0.5	0.3 ± 0.1	—
C5 adaptive lmin0.05	0.922 ± 0.001	0.023 ± 0.004	0.930 ± 0.005	0.016 ± 0.002	0.802 ± 0.009	0.921 ± 0.001	2.6 ± 0.5	0.1 ± 0.0	—
C6 adaptive lmin0.10	0.927 ± 0.002	0.028 ± 0.005	0.926 ± 0.004	0.017 ± 0.003	0.806 ± 0.009	0.927 ± 0.002	3.0 ± 0.4	0.0 ± 0.0	—
C7 adaptive lmin0.20	0.931 ± 0.002	0.035 ± 0.005	0.919 ± 0.003	0.021 ± 0.004	0.812 ± 0.009	0.930 ± 0.001	3.7 ± 0.4	0.0 ± 0.0	—

Table C.2: rot-MNIST λ-curriculum sweep, full metrics, at both momentum settings. All conditions applied on top of standard ER (1 k reservoir).

C.3 rot-MNIST: Buffer-Fidelity Diagnostics

The gradient diagnostics of Appendix B were logged for the decomposition block only. The full-data conditions (D2, D4) reach exact fidelity by construction, the replay gradient equals the deterministic empirical past-task gradient, so cosine and magnitude ratio are 1.000. The finite-buffer conditions (D1, D3) show the directional and magnitude error a 1,000-sample reservoir incurs against g_{true}.

Condition	$\bar{\cos}$	$\cos_{\min}$	$\bar{\rho}$
D1 vanilla (μ=0)	0.711 ± 0.035	0.086 ± 0.103	1.040 ± 0.039
D3 balanced (μ=0)	0.665 ± 0.043	0.025 ± 0.140	1.137 ± 0.060
D1 vanilla (μ=0.9)	0.551 ± 0.033	0.135 ± 0.069	0.418 ± 0.014
D3 balanced (μ=0.9)	0.441 ± 0.039	-0.052 ± 0.117	0.265 ± 0.024
D2, D4 (full data)	1.000	1.000	1.000

Table C.3: Buffer-fidelity diagnostics for the decomposition block: $\bar{\cos}$ **and** $\cos_{\min}$ **are the mean and minimum of** $\cos(g_{\mathbf{replay}}, g_{\mathbf{true}})$ **over the window;** $\bar{\rho}$ **is the mean of** $\|g_{\mathbf{replay}}\|/\|g_{\mathbf{true}}\|$**.**

C.4 CIFAR-10: Headline Block

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end
G1 vanilla BN	0.723 ± 0.008	0.010 ± 0.019	0.644 ± 0.023	0.127 ± 0.015	0.484 ± 0.020	0.667 ± 0.016	11.8 ± 2.5	17.5 ± 4.6
G2 NCL BN	0.710 ± 0.005	0.064 ± 0.026	0.504 ± 0.045	0.167 ± 0.031	0.513 ± 0.018	0.611 ± 0.020	25.8 ± 5.2	55.1 ± 11.2
G3 adaptive BN	0.722 ± 0.011	0.006 ± 0.016	0.713 ± 0.017	0.070 ± 0.010	0.469 ± 0.016	0.698 ± 0.013	4.8 ± 0.7	3.6 ± 2.2
G4 vanilla GN	0.730 ± 0.011	-0.023 ± 0.030	0.309 ± 0.111	0.294 ± 0.097	0.415 ± 0.058	0.507 ± 0.054	41.2 ± 13.5	34.9 ± 23.7
G5 NCL GN	0.711 ± 0.017	0.012 ± 0.028	0.423 ± 0.083	0.200 ± 0.062	0.391 ± 0.055	0.563 ± 0.038	29.7 ± 10.0	52.6 ± 13.3
G6 adaptive GN	0.731 ± 0.020	-0.026 ± 0.026	0.670 ± 0.055	0.095 ± 0.033	0.340 ± 0.052	0.687 ± 0.037	5.2 ± 2.1	1.7 ± 0.9

Table C.4: CIFAR-10 headline block, full metrics (ResNet-18, momentum 0.9**, buffer** $5{,}000$**, five seeds). "adaptive" is the shipped curriculum with** $\lambda_{\min} = 0.20$**.**

C.5 CIFAR-10: Generalisation Block

Each shift pairs the shipped curriculum ("ship": adaptive $\lambda_{\min} = 0.20$, buffer 5,000) against a vanilla-ER control on the same shift, at the indicated normalisation, over three seeds.

Condition	ACC	FORG	min-ACC	WC-ACC	Depth	Area end
BatchNorm						
shot noise — vanilla	0.724 ± 0.011	0.017 ± 0.010	0.633 ± 0.017	0.663 ± 0.014	13.1 ± 3.0	21.2 ± 3.9
shot noise — ship	0.723 ± 0.014	0.010 ± 0.010	0.708 ± 0.032	0.697 ± 0.016	5.6 ± 1.0	5.8 ± 1.5
contrast — vanilla	0.671 ± 0.026	-0.008 ± 0.006	0.610 ± 0.063	0.586 ± 0.048	15.4 ± 3.9	33.4 ± 4.5
contrast — ship	0.671 ± 0.044	-0.012 ± 0.007	0.727 ± 0.034	0.643 ± 0.048	3.7 ± 1.1	2.2 ± 0.9
3-task — vanilla	0.740 ± 0.006	-0.011 ± 0.013	0.670 ± 0.015	0.691 ± 0.010	3.3 ± 0.4	12.0 ± 2.4
3-task — ship	0.745 ± 0.010	-0.021 ± 0.008	0.677 ± 0.025	0.696 ± 0.017	6.9 ± 2.7	4.1 ± 1.1
GroupNorm						
shot noise — vanilla	0.718 ± 0.021	-0.036 ± 0.036	0.244 ± 0.075	0.469 ± 0.032	46.3 ± 12.0	54.4 ± 40.8
shot noise — ship	0.705 ± 0.061	-0.028 ± 0.012	0.638 ± 0.108	0.656 ± 0.087	7.3 ± 5.9	2.4 ± 1.1
contrast — vanilla	0.777 ± 0.022	-0.067 ± 0.021	0.641 ± 0.061	0.710 ± 0.040	8.4 ± 4.4	3.3 ± 0.6
contrast — ship	0.778 ± 0.032	-0.070 ± 0.013	0.668 ± 0.077	0.724 ± 0.054	3.9 ± 3.1	1.2 ± 0.8
3-task — vanilla	0.722 ± 0.020	-0.023 ± 0.015	0.468 ± 0.043	0.550 ± 0.023	4.4 ± 0.4	24.8 ± 11.8
3-task — ship	0.721 ± 0.019	-0.025 ± 0.020	0.421 ± 0.114	0.517 ± 0.070	32.8 ± 14.5	16.9 ± 10.2

Table C.5: CIFAR-10 generalisation block, full metrics. The two-task novel corruptions (shot noise, contrast) favour the curriculum under both normalisations; the three-task group-norm sequence is the outlier discussed in Section 7.5.

C.6 Long-Sequence rot-MNIST (L-series)

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end
Momentum 0.0								
L1 vanilla ER	0.861 ± 0.004	0.060 ± 0.004	0.670 ± 0.055	0.207 ± 0.053	0.443 ± 0.011	0.722 ± 0.043	21.0 ± 5.8	3.14 ± 0.77
L2 curriculum $\lambda_{\min}$=0.20	0.860 ± 0.002	0.031 ± 0.005	0.828 ± 0.007	0.073 ± 0.005	0.400 ± 0.003	0.840 ± 0.005	4.1 ± 0.8	1.62 ± 0.42
L3 asym. PER δ=0.1	0.870 ± 0.003	0.053 ± 0.006	0.843 ± 0.008	0.038 ± 0.007	0.381 ± 0.003	0.862 ± 0.007	4.7 ± 1.2	0.38 ± 0.24
Momentum 0.9								
L1 vanilla ER	0.889 ± 0.007	0.096 ± 0.008	0.845 ± 0.007	0.065 ± 0.008	0.317 ± 0.006	0.870 ± 0.006	8.9 ± 1.7	1.68 ± 0.69
L2 curriculum $\lambda_{\min}$=0.20	0.897 ± 0.002	0.071 ± 0.003	0.871 ± 0.005	0.041 ± 0.004	0.364 ± 0.005	0.888 ± 0.004	5.4 ± 0.6	0.60 ± 0.14
L3 asym. PER δ=0.1	0.884 ± 0.004	0.104 ± 0.004	0.849 ± 0.006	0.054 ± 0.004	0.325 ± 0.005	0.873 ± 0.005	7.5 ± 1.5	0.80 ± 0.63

Table C.6: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive transitions, five seeds), the refactored L-series: the two shipping gates against vanilla ER, at both momentum settings. Depth is the worst single-boundary drop (`gap_depth`) and area the end-referenced transient (`gap_area_end`), aggregated as in the two-task blocks so the numbers are directly comparable. At momentum 0 both gates cut the depth from 21 pp to ≈4 pp; under momentum, where the per-boundary drops are already smaller, the curriculum stays ahead on cumulative retention (lowest FORG, highest ACC and WC-ACC) while asymmetric PER holds the smallest per-boundary area but, its per-step filter re-inflated by momentum (Section 7.3), gives up its retention edge and tracks vanilla on FORG.

C.7 Supplementary Figures

This subsection collects supplementary per-step curves: the five-task rot-MNIST sequence (the multi-transition check reported in Table C.6) and the three-task group-norm generalisation outlier discussed in Section 7.5.

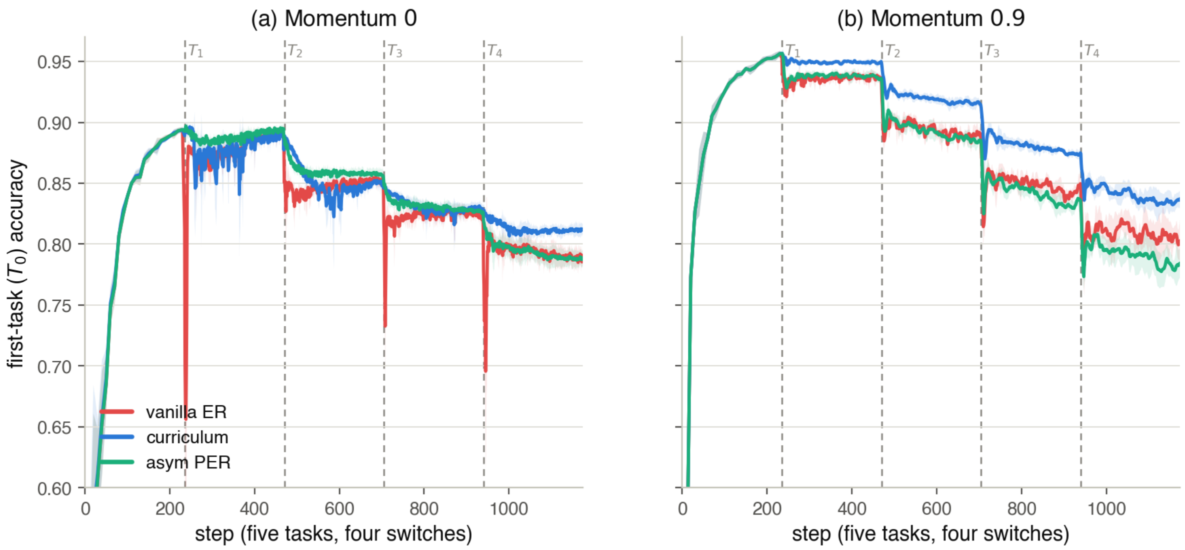

Figure C.1: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive switches, five seeds with $\pm$std bands), the refactored L-series: first-task T_0 accuracy across the whole sequence for vanilla ER (red), the shipping curriculum $\lambda_{\min}$=0.20 (blue) and asymmetric PER δ=0.1 (aqua); dashed lines mark the four switches, and the two panels are the two momentum settings. (a) Momentum 0: both gates hold T_0 almost flat through every boundary — asymmetric PER the smoothest of all, with only shallow dips (smallest per-boundary area, Table C.6) — while vanilla ER takes a sharp instantaneous drop at each switch. This is the near-gap-free directional-gate regime the two-task sweep also finds at momentum 0. (b) Momentum 0.9: momentum re-inflates PER's per-step filter, exactly as on the two-task benchmark (Section 7.3), so its boundary transients return and it tracks vanilla on cumulative retention; the curriculum keeps the boundaries smoothest and holds the highest plateau, the least cumulative forgetting on the long horizon. The single-transition reading thus carries past one boundary, and the momentum roles carry with it.

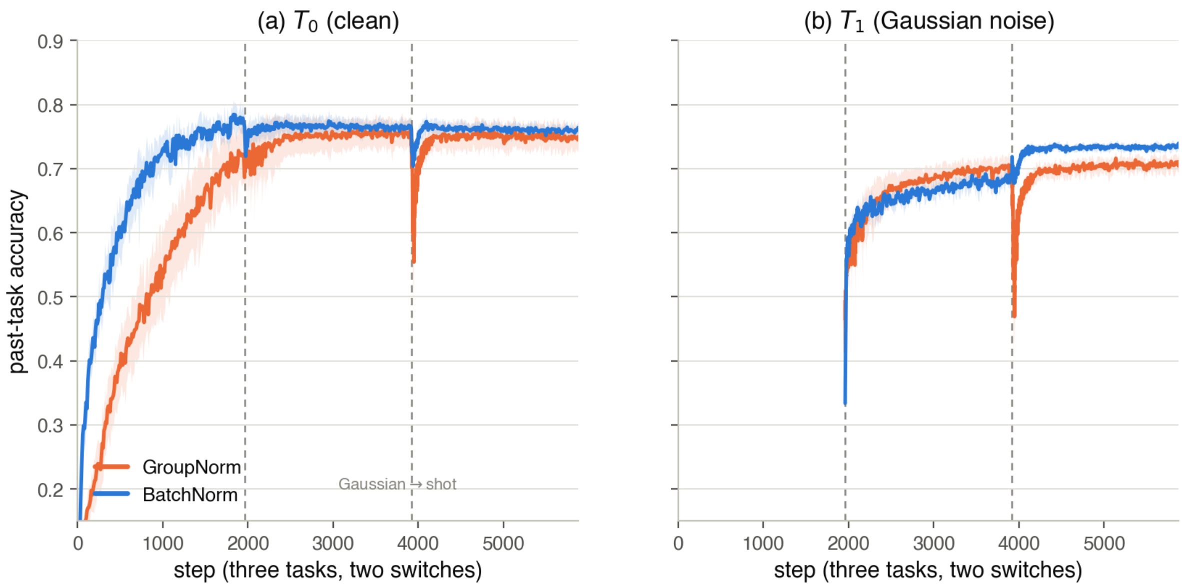

Figure C.2: Per-step past-task accuracy across the three-task CIFAR-10 generalisation sequence (none → Gaussian → shot, ResNet-18, shipping curriculum, three seeds with ±std bands), comparing the group-norm run (orange) against its batch-norm counterpart (blue). The two switches (dashed) fall at ∼2k and ∼4k steps. The first boundary (none → Gaussian) barely perturbs the past tasks in either norm. The second (Gaussian → shot, two near-identical corruptions) drives a deep transient in T_0 (a) and T_1 (b) under group norm only, while the batch-norm run holds flat through both switches. The dip is thus localised entirely to the second transition and to group norm, the adaptive-ratio reliability boundary discussed in Section 7.5.

D Detailed Step-by-Step Derivation of the Trajectory Bound

The Mathematical Setup & Definitions

1. The Combined Loss. The total loss function at any point $\lambda \in [0, 1]$ in the curriculum is defined as:

$$L_\lambda(\theta) = L_{\text{replay}}(\theta) + \lambda \cdot L_{\text{current}}(\theta) \tag{D.1}$$

2. Defining the Hessians (∇^2). When we take the derivative of a gradient (∇), we obtain the second derivative matrix, called the Hessian (H).

- $H_{\text{replay}} = \nabla^2 L_{\text{replay}}(\theta)$ (The curvature of the old task)
- $H_{\text{current}} = \nabla^2 L_{\text{current}}(\theta)$ (The curvature of the new task)

Therefore, the Hessian of our combined loss, H_λ, is defined by applying the second derivative operator to the total loss function:

$$\begin{aligned} H_\lambda &= \nabla^2 \left[L_{\text{replay}}(\theta) + \lambda \cdot L_{\text{current}}(\theta) \right] \\ H_\lambda &= \nabla^2 L_{\text{replay}}(\theta) + \lambda \cdot \nabla^2 L_{\text{current}}(\theta) \\ H_\lambda &= H_{\text{replay}} + \lambda \cdot H_{\text{current}} \end{aligned}$$

3. Assumption: Local-Minimum along the Interpolation Branch. We assume that the loss functions L_{replay} and L_{current} are twice continuously differentiable (C^2) in a neighbourhood of the interpolation path. Furthermore, we assume there exists a continuously differentiable (C^1) branch of critical points $\lambda \mapsto \Theta^*(\lambda)$ for the interpolated loss L_λ, initialized at $\Theta^*(0) = \theta_0^*$, such that the Hessian remains strictly positive-definite: $H_\lambda(\Theta^*(\lambda)) \succ 0$ for every $\lambda \in [0, 1]$.

This assumption ensures that the iterate tracks a continuous family of strict local minima rather than relying on global convexity. Near $\lambda = 0$, this condition is naturally satisfied. Because the initial state θ_0^* is a non-degenerate local minimum of L_{replay}, it follows that $H_0 = H_{\text{replay}}(\theta_0^*) \succ 0$. Consequently, the implicit function theorem guarantees a unique smooth branch in the immediate neighbourhood of $\lambda = 0$.

The persistence of this positive-definite branch as λ increases is bounded by Weyl's inequality applied to $H_\lambda = H_{\text{replay}} + \lambda H_{\text{current}}$:

$$\lambda_{\min}(H_\lambda) \;\geq\; \lambda_{\min}(H_{\text{replay}}) + \lambda \, \lambda_{\min}(H_{\text{current}}).$$

If the new task is locally convex at these points ($H_{\text{current}} \succeq 0$), positive-definiteness holds for all $\lambda \geq 0$ without requiring a global convexity assumption.

Part A: The Perfect Path Never Spikes

Let $\Theta^*(\lambda)$ be the "Optimum Path", the exact parameter state that perfectly minimises $L_\lambda(\theta)$ for any given λ.

Step 1: The First-Order Condition. Because $\Theta^*(\lambda)$ is the perfect minimum, the gradient (slope) of the total loss evaluated at that exact point must be zero. We define this as our starting equilibrium:

$$\begin{aligned} \nabla \left[L_\lambda(\Theta^*(\lambda)) \right] &= 0 \\ \nabla \left[L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot L_{\text{current}}(\Theta^*(\lambda)) \right] &= 0 \end{aligned}$$

Distributing the gradient operator to both terms gives us the fundamental First-Order Condition:

$$\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0 \tag{D.2}$$

Step 2: Implicit Differentiation to Find Path Movement. We want to know how the optimal path Θ^* moves as λ changes. We apply the derivative operator $\frac{d}{d\lambda}$ to the entire First-Order Condition:

$$\frac{d}{d\lambda}\left[\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\right] = \frac{d}{d\lambda}[0]$$

Distributing the operator to each term:

$$\frac{d}{d\lambda}\left[\nabla L_{\text{replay}}(\Theta^*(\lambda))\right] + \frac{d}{d\lambda}\left[\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\right] = 0$$

Evaluating the first term via the multivariable chain rule:

$$\frac{d}{d\lambda}\left[\nabla L_{\text{replay}}(\Theta^*(\lambda))\right] = \nabla^2 L_{\text{replay}}(\Theta^*(\lambda)) \cdot \frac{d\Theta^*}{d\lambda}$$

Evaluating the second term via the product rule and chain rule:

$$\frac{d}{d\lambda}\left[\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\right] = (1) \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) + \lambda \cdot \left(\nabla^2 L_{\text{current}}(\Theta^*(\lambda)) \cdot \frac{d\Theta^*}{d\lambda}\right)$$

Substituting these expanded terms back into the equation:

$$\left[\nabla^2 L_{\text{replay}}(\Theta^*) \cdot \frac{d\Theta^*}{d\lambda}\right] + \left[\nabla L_{\text{current}}(\Theta^*) + \lambda \cdot \nabla^2 L_{\text{current}}(\Theta^*) \cdot \frac{d\Theta^*}{d\lambda}\right] = 0$$

Grouping the terms that share the path movement $\frac{d\Theta^*}{d\lambda}$:

$$\left(\nabla^2 L_{\text{replay}}(\Theta^*) + \lambda \cdot \nabla^2 L_{\text{current}}(\Theta^*)\right) \cdot \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}}(\Theta^*) = 0$$

Recognising the definition of the combined Hessian H_λ inside the parentheses:

$$H_\lambda \cdot \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0$$

Solving algebraically for the path movement $\frac{d\Theta^*}{d\lambda}$:

$$\frac{d\Theta^*}{d\lambda} = -H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda)) \tag{D.3}$$

Step 3: Calculating the Change in Task 0 Loss. How does the Replay loss specifically change as λ increases along this perfect path? We apply the derivative operator:

$$\frac{d}{d\lambda}\left[L_{\text{replay}}(\Theta^*(\lambda))\right] = \nabla L_{\text{replay}}(\Theta^*(\lambda))^\top \cdot \frac{d\Theta^*(\lambda)}{d\lambda}$$

We perform two substitutions here. First, from (D.2), we know $\nabla L_{\text{replay}} = -\lambda \cdot \nabla L_{\text{current}}$. Second, from (D.3), we substitute the path movement.

$$\begin{aligned}\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) &= \left(-\lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\right)^\top \cdot \left(-H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\right) \\ &= \lambda \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))^\top \cdot H_\lambda^{-1} \cdot \nabla L_{\text{current}}(\Theta^*(\lambda))\end{aligned}$$

Because H_λ is strictly positive-definite along the followed branch of local minima for all $\lambda \in [0, 1]$ (per the Local-Minimum Assumption), its inverse H_λ^{-1} is also positive-definite. The quadratic form $v^\top M v$ is therefore guaranteed to be non-negative. Since $\lambda \geq 0$, the entire equation satisfies:

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) \geq 0$$

Conclusion for Part A: The Replay loss mathematically never decreases along the perfect path, meaning it can never overshoot its final joint-optimum value.

Part B: The Clumsy Network Tracks the Perfect Path

Let $\theta(t)$ be the actual position of the model parameters at time t. We define the tracking error $\delta(t)$:

$$\delta(t) = \theta(t) - \Theta^*(\lambda(t))$$

Step 1: Differentiating the Error. We apply the time derivative operator $\frac{d}{dt}$ to our error definition:

$$\frac{d}{dt}[\delta(t)] = \frac{d}{dt}[\theta(t)] - \frac{d}{dt}[\Theta^*(\lambda(t))]$$

The model moves via gradient descent, so $\frac{d\theta}{dt} = -\nabla L_{\lambda(t)}(\theta(t))$. The perfect path moves via the chain rule $\frac{d\Theta^*}{dt} = \frac{d\Theta^*}{d\lambda} \cdot \frac{d\lambda}{dt}$. Because the curriculum has a constant velocity over N steps, $\frac{d\lambda}{dt} = \frac{1}{N}$.

$$\frac{d\delta}{dt} = -\nabla L_{\lambda(t)}(\theta(t)) - \left(\frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N}\right)$$

Step 2: Taylor Expansion of the Actual Gradient. We approximate the gradient at the actual position $\theta(t)$ by Taylor-expanding it around the perfect position Θ^*:

$$\nabla L_\lambda(\theta) \approx \nabla L_\lambda(\Theta^*) + \nabla\left[\nabla L_\lambda(\Theta^*)\right] \cdot (\theta - \Theta^*)$$

Since the gradient at the perfect minimum is zero ($\nabla L_\lambda(\Theta^*) = 0$) and the derivative of the gradient is the Hessian ($\nabla^2 L_\lambda = H_\lambda$), this collapses to:

$$\nabla L_\lambda(\theta) \approx H_\lambda \cdot \delta$$

Step 3: The Final Steady State. Substitute this expanded gradient back into our differential equation:

$$\frac{d\delta}{dt} \approx -H_\lambda \cdot \delta - \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N}$$

Assuming the lag reaches a steady state, the error stops growing ($\frac{d\delta}{dt} \approx 0$). This is a first-order, gradient-flow approximation: it linearises the gradient around Θ^* and treats the dynamics as continuous, so it captures the scaling of the lag with N rather than an exact discrete-time bound.

$$\begin{aligned} 0 &\approx -H_\lambda \cdot \delta - \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \\ H_\lambda \cdot \delta &\approx -\frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \\ \delta &\approx -H_\lambda^{-1} \cdot \frac{d\Theta^*}{d\lambda} \cdot \frac{1}{N} \end{aligned}$$

Taking the norm, we see the lag distance is mathematically bounded by the curriculum length N:

$$\|\delta\| = O\left(\frac{1}{N}\right) \tag{D.4}$$

Combining Part A and B: Expanding the Actual Loss

To find the maximum Replay loss the model will actually experience, we Taylor expand the loss of the actual trajectory around the perfect position:

$$\begin{aligned} L_{\text{replay}}(\theta) &\approx L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top \cdot (\theta - \Theta^*) \\ L_{\text{replay}}(\theta) &\approx L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top \cdot \delta \end{aligned}$$

From Part A, $L_{\text{replay}}(\Theta^*)$ cannot exceed the final joint optimum loss $L_{\text{replay}}(\theta^*_{\text{joint}})$. From Part B, the lag δ is $O(1/N)$. Substituting these limits yields the final bound:

$$\max_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta^*_{\text{joint}}) + O\left(\frac{1}{N}\right) \tag{D.5}$$

Conclusion: The transient stability gap is entirely contained within the $O(1/N)$ term, meaning a sufficiently slow continuous curriculum eliminates the discrete-time overshoot.